Circle

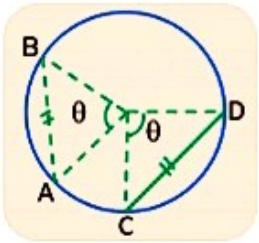
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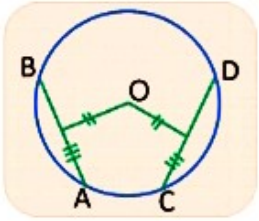
Circle

Basic Geometry with circles

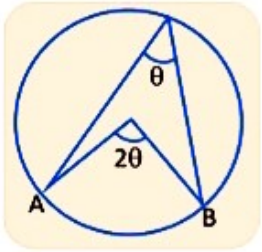
1. Equal chords subtend equal angles at the center and vice-versa.



2. Equal chords of a circle are equidistant from the center and vice-versa.



3. Angle subtended by an arc at the center is double the angle subtended at any point on the remaining part of the circle

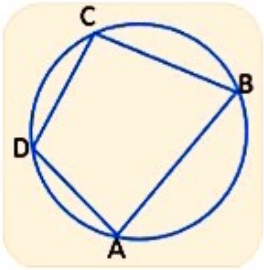


4. Angles in the same segment of circle are equal.



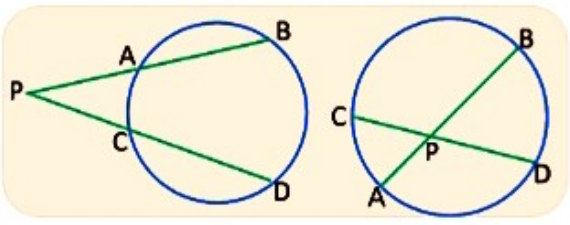
5. The sum of the opposite angles of a cyclic quadrilateral is 180° and vice-versa.

$$\angle A + \angle C = \angle B + \angle D = 180°$$



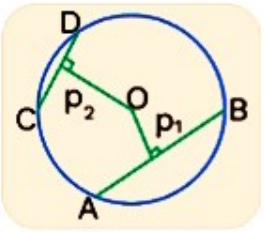
6. If two chords of a circle intersect either inside or outside the circle, the rectangle contained by the parts of one chord is equal in area to rectangle by the parts of other.

$$PA \times PB = PC \times PD$$



7. The greater of the two chords in a circle is nearer to the centre than lesser.

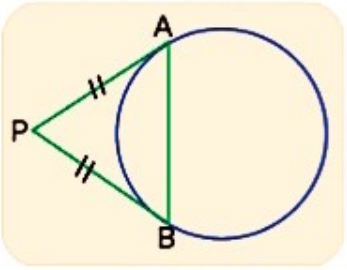
$\because$ AB > CD, $\therefore$ $p_1 < p_2$



8. A chord drawn across the circular region divides it in two parts each of which is called a segment of the circle

9. The tangents at the extremities of a chord of a circle are equal.

PA = PB



Circle

Definition

Circle is defined as a locus of a point 'P' which moves in x – y plane in such a way that its distance from the fixed point in the same plane is always constant.

Standard Form

$(x - a)^2 + (y - b)^2 = r^2$

(a, b) ≡ center

r ≡ radius

General Equation of the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$

center ≡ (–g, –f)

$= \left(-\frac{1}{2} \text{ coefficient of } x, -\frac{1}{2} \text{coefficient of } y\right)$

Radius = $\sqrt{g^2 + f^2 - c}$

Example :

Q **Find equation of circle whose radius is 3 and center is (–1, 2).**

Sol. Equation is

$(x + 1)^2 + (y - 2)^2 = 3^2$

Q **Find equation of circle whose radius is 10 and center is (–5, –6).**

Sol. Equation is

$(x + 5)^2 + (y + 6)^2 = 10^2$

Necessary and sufficient condition for General Equation of 2nd degree to represent a circle.

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

(a) Coefficient of x^2 = coefficient of y^2 (not necessarily unity) and

(b) Coefficient of $xy = 0$

Note :

The general equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ contains 3 independent arbitrary constants g, f and c which means that a unique circle passes through 3 non-collinear points. Hence 3 points on a circle must be given to determine the unique equation of the circle.

Nature of circle :

$(x^2 + y^2) + 2gx + 2fy + c = 0$

(i) If $g^2 + f^2 - c > 0$

$\Rightarrow$ **Real circle with finite radius**

(ii) If $g^2 + f^2 - c = 0$

$\Rightarrow$ **Point circle**

(iii) If $g^2 + f^2 - c < 0$

$\Rightarrow$ **Imaginary circle**

Example :

Q **Find the equation of the circle passing through the points (3, 4), (–3, –4), (0, 5).**

Sol. Let equation of circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$

Now on satisfying the equation by given points,

$\Rightarrow 9 + 16 + 6g + 8f + c = 0$

$6g + 8f + c = -25$ ...(1)

$\Rightarrow 9 + 16 - 6g - 8f + c = 0$

$6g + 8f - c = 25$ ...(2)

$\Rightarrow 0 + 25 + 0 + 10f + c = 0$

$10f + c = -25$ ...(3)

Subtracting (2) from (1)
$\Rightarrow 2c = -50 \Rightarrow c = -25$
from (3) $\Rightarrow 10f = -25 - c \Rightarrow f = 0$
from (2) $\Rightarrow 6g + 0 + 25 = 25 \Rightarrow g = 0$
Hence, equation is $x^2 + y^2 - 25 = 0$

Alternate Method :
For ΔABC, A(3, 4), B(−3, −4) , C(0, 5) origin will be circumcentre which is centre of circle passing through A, B and C and radius $r = OA = \sqrt{3^2 + 4^2} = 5$
equation $(x - 0)^2 + (y - 0)^2 = 5^2$

Q **Find the equation of the circle having lines $2x - 3y = 5$ and $3x - 4y = 7$ as its diameter / normal / longest chord and whose area is 154 sq. units.**

Sol. Centre will be point of intersection of the diameters
$d_1 : 2x - 3y = 5$ ...(1)
$d_2 : 3x - 4y = 7$ ...(2)
$(2) \times 3 - (1) \times 4$
$x = 1, y = -1 \Rightarrow$ centre $(1, -1)$
Area $= 154 \Rightarrow \pi r^2 = 154$

$$r^2 = \frac{154}{\pi} = \frac{154}{22} \times 7 = 49$$

Hence, equation is : $(x - 1)^2 + (y + 1)^2 = 49$

Q **Find the equation of the circumcircle of Δ formed by the lines $xy + 2x + 2y + 4 = 0$, $x + y + 2 = 0$**

Sol. $xy + 2x + 2y + 4 = x(y + 2) + 2(y + 2)$
$= (x + 2)(y + 2)$
Sides of triangle $L_1 : x + 2 = 0$
$L_2 : y + 2 = 0$
$L_3 : x + y + 2 = 0$
$\because L_1 \perp L_2$ hence circumcentre will be mid point of hypotenuse
L_1 and $L_3 \Rightarrow A(-2, 0)$, L_2 and $L_3 \Rightarrow C(0, -2)$

$$\text{Centre : } P\left(\frac{-2 + 0}{2}, \frac{0 - 2}{2}\right) = P(-1, -1)$$

$$\text{radius} = \frac{AC}{2} = \frac{\sqrt{4 + 4}}{2} = \sqrt{2}$$

Hence, equation is : $(x + 1)^2 + (y + 1)^2 = 2$

Q **Find centre and radius of the circle** $2x^2 + 2y^2 - 6x + 8y - 5 = 0$.

Sol. Given equation is

$$x^2 + y^2 - 3x + 4y - \frac{5}{2} = 0$$

$$\text{centre} \equiv \left(-\frac{1}{2}(-3), -\frac{1}{2}(4)\right) \equiv \left(\frac{3}{2}, -2\right)$$

$$\text{radius} = \sqrt{\left(-\frac{3}{2}\right)^2 + (2)^2 + \frac{5}{2}}$$

$$\sqrt{\frac{9}{4} + 4 + \frac{5}{2}} = \frac{\sqrt{35}}{2} \text{ unit}$$

Q **Find equation of circle concentric with $3x^2 + 3y^2 - 5x - 6y - 14 = 0$ and perimeter of its semicircle is 36.**

Sol. Given circle is $x^2 + y^2 - \frac{5}{3}x - 2y - \frac{14}{3} = 0$

Centre: $\left(\frac{5}{6}, 1\right)$

perimeter of semicircle, $2r + \pi r = 36$

$$r\left(2 + \frac{22}{7}\right) = 36 \Rightarrow r = 7$$

Hence, equation of required circle is $\left(x - \frac{5}{6}\right)^2 + (y - 1)^2 = 49$

Q **Find equation of the circle which passes through (2, 3) and centre on the x-axis, radius being 5.**

Sol. Let centre be $(\alpha, 0)$, given $r = 5$
Equation will be $(x - \alpha)^2 + (y - 0)^2 = 5^2$
which passes through (2, 3) hence
$(2 - \alpha)^2 + 9 = 25$
$(2 - \alpha)^2 = 16 \Rightarrow 2 - \alpha = 4, -4$
$\alpha = -2, 6$
Hence, equations are $(x + 2)^2 + y^2 = 25$
or $(x - 6)^2 + y^2 = 25$

Q **Find the equation of the circle for which centre is on the line y = 2x and circle passes through (–1, 2) and (3, –2).**

Sol. Perpendicular bisector of AB where A (–1, 2) and B (3, –2) will be another diameter

Hence, mid-point will be, (1, 0)

slope $m_{AB} = \frac{4}{-4} = -1$

Equation will be $(y - 0) = 1\,(x - 1) \Rightarrow y = x - 1$ ...(1)

and $y = 2x$ ...(2)

from (1) and (2)

centre O (–1, –2)

Radius = OA = $\sqrt{0^2 + 4^2} = 4$

Hence, equation is : $(x + 1)^2 + (y + 2)^2 = 4^2$

Q **A circle is drawn with its centre on the line x + y = 2 to touch the line 4x – 3y + 4 = 0 and pass through the point (0, 1). Find its equation.**

Sol. Let centre be O(α, β), A(0, 1)

which lies on x + y = 2

$\Rightarrow \alpha + \beta = 2$ ...(1)

Now radius is length of perpendicular from O to line 4x – 3y + 4 = 0

(as line of 2 tangent to circle)

Hence, $\left|\frac{4\alpha - 3\beta + 4}{5}\right|$ = OA (radius)

$$|4\alpha - 3\beta + 4| = 5\sqrt{\alpha^2 + (\beta - 1)^2}$$

$$|4\alpha - 3(2 - \alpha) + 4| = 5\sqrt{\alpha^2 + (2 - \alpha - 1)^2}$$

$$|7\alpha - 2| = 5\sqrt{2\alpha^2 - 2\alpha + 1}$$

On squaring,

$$49\alpha^2 - 28\alpha + 4 = 25(2\alpha^2 - 2\alpha + 1)$$

$$\alpha^2 - 22\alpha + 21 = 0$$

$$(\alpha - 21)(\alpha - 1) = 0$$

$\Rightarrow \alpha = 1, 21$

$\beta = 1, -19$

Centre will be (1, 1) or (21, –19)

for centre (1, 1), radius = $\sqrt{1^2 + 0^2} = 1$

Equation is: $(x-1)^2+(y-1)^2=1$

for centre (21, –19), radius$=\sqrt{21^2+20^2}=\sqrt{841}=29$

equation : $(x-21)^2+(y+19)^2=29^2$

Q **Locus of the image of the point (2, 3) in the line $(2x-3y+4)+k(x-2y+3)=0, k\in R$ is a**

(A) Circle of radius $\sqrt{3}$ **(B) Straight line parallel to X – axis**

(C) Straight line parallel to Y – axis **(D) Circle of radius $\sqrt{2}$**

Sol. **(D)**

Given family of straight line

$L_1+kL_2=0$

will always pass-through point of intersection of L_1 and L_2

$L_1: 2x-3y+4=0$ ...(1)

$L_2: x-2y+3=0$ ...(2)

(1)–(2) × 2

$y=2, x=1$ these point A(1, 2)

Now let image of B(2, 3) is P(h, k) then perpendicular bisector of BP will pass through A

hence mid-point $M\left(\frac{h+2}{2},\frac{k+3}{2}\right)$

slope $=m_{BP}=\frac{k-3}{h-2}$

Equation is: $\left(y-\frac{k+3}{2}\right)=-\left(\frac{h-2}{k-3}\right)\left(x-\frac{h+2}{2}\right)$

which passes through A hence

$$\left(2-\frac{k+3}{2}\right)=-\left(\frac{h-2}{k-3}\right)\left(1-\frac{h+2}{2}\right)$$

$$\frac{1-k}{2}=-\frac{(h-2)}{(k-3)}\left(\frac{-h}{2}\right)$$

$$k-k^2-3+3k=h^2-2h$$

$\Rightarrow h^2+k^2-2h-4k+3=0$

Locus: $x^2+y^2-2x-4y+3=0$

centre (1, 2), radius $=\sqrt{2}$

Q **Find the equation of circle whose centre is (4, 3) and touches the line $5x - 12y - 10 = 0$.**

Sol. Centre C(4, 3)
Radius = Perpendicular distance of C from given tangent

$$r = \left|\frac{5(4) - 12(3) - 10}{13}\right| = 2$$

Equation is : $(x-4)^2 + (y-3)^2 = 4$

Diametrical Form of Circle

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Where (x_1, y_1) and (x_2, y_2) are diametrical opposite ends.

Examples :

Q **Find the equation of the circle of least radius passing through the points. (2, 3), (3, 1).**

Sol. Given point will be diametrical end points of required circle
Equation: $(x - 2)(x - 3) + (y - 3)(y - 1) = 0$

$$(x^2 + y^2) - 5x - 4y + 9 = 0$$

Q **Find equation of tangent to circle parallel to tangent $x + y = 5$, where centre of the circle is (1, 2).**

Sol. Let tangent is $x + y = \lambda$
Since both are tangents hence perpendicular distance of tangent from centre will be radius

$$\Rightarrow \left|\frac{1 + 2 - \lambda}{\sqrt{2}}\right| = \left|\frac{1 + 2 - 5}{\sqrt{2}}\right|$$

$$\Rightarrow |3 - \lambda| = 2 \Rightarrow \lambda = 5, 1$$

Equation: $x + y = 1$

Q **The abscissa of 2 points 'A' and 'B' are the roots of the equation $x^2 + 2x - 3 = 0$ and the ordinate are the roots of the equation $y^2 - 4y + 1 = 0$.**

Find the equation of circle AB as diameter.

Sol. Let $A(x_1, y_1)$ and $B(x_2, y_2)$

Now given

x_1, x_2 are roots of $x^2 + 2x - 3 = 0$

$\Rightarrow x^2 + 2x - 3 = (x - x_1)(x - x_2)$ ...(1)

y_1, y_2 are roots of $y^2 - 4y + 1 = 0$

$\Rightarrow y^2 - 4y + 1 = (y - y_1)(y - y_2)$ ...(2)

Equation of circle diameter AB is

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

$x^2 + y^2 + 2x - 4y - 2 = 0$ {from (1) and (2)}

Q **Find the equation of the circle which touches the line x = 0, y = 0 and x = 4?**

Sol. Let radius = r

Now, Diameter = distance between x = 0, x = 4

$2r = 4 \Rightarrow r = 2$

Centre will be $\left(\frac{0+4}{2}, \pm r\right) \equiv (2, 2)$ or $(2, -2)$

Equation : $(x - 2)^2 + (y \pm 2)^2 = 2^2$

Q **Line y = mx + c cuts the curve $y^2 = 4ax$ at A and B. Find the equation of circle with AB as diameter.**

Sol. Let $A(x_1, y_1)$ and $B(x_2, y_2)$

$y = mx + c$ ...(1)

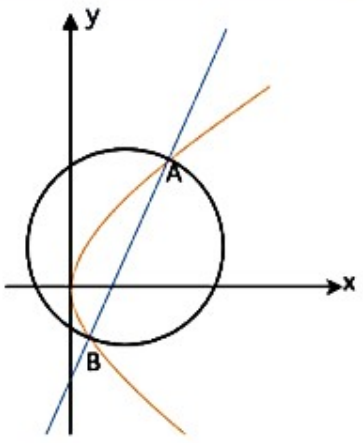
and $y^2 = 4ax$ ...(2)

by (1) and (2),

$(mx + c)^2 = 4ax$ → x_1, x_2

$m^2x^2 + (2mc - 4a)x + c^2 = m^2(x - x_1)(x - x_2)$...(3)2

Again by (1) and (2)



$$y^2 = 4a\left(\frac{y - c}{m}\right)$$

$\Rightarrow my^2 - 4ay + 4ac = 0$ → y_1, y_2

$my^2 - 4ay + 4ac = m(y - y_1)(y - y_2)$ ...(4)

Equation: $m^2 (x - x_1)(x - x_2) + m^2(y - y_1)(y - y_2) = 0$

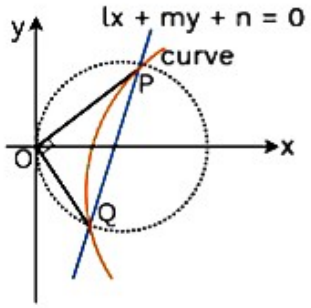
$m^2x^2 + m^2y^2 + (2mc - 4a)x - 4amy + c^2 + 4acm = 0$

Q **The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2 (a + b) = l^2 + m^2$.**

Sol. Equation of pair of Straight lines by homogenization

$$ax^2 + 2hxy + by^2 = \left(\frac{lx + my}{-n}\right)^2$$

$$\left(n^2a - l^2\right)x^2 + \left(2hn^2 - 2lm\right)xy + \left(n^2b - m^2\right)y^2 = 0$$



Which must be perpendicular
(∵ circle with PQ diameter passes through origin)

$\Rightarrow$ coeff. of x^2 + coeff. of $y^2 = 0$

$$\left(n^2a - l^2\right) + \left(n^2b - m^2\right) = 0$$

$$n^2(a + b) = l^2 + m^2$$

Hence proved.

Q **A rectangle is inscribed in a circle with a diameter lying along the line, $3y = x + 7$. If the two adjacent vertices of the rectangle are $(-8, 5)$ and $(6, 5)$ then the area of the rectangle (in sq. units) is:**

(A) 72 **(B) 84** **(C) 98** **(D) 56**

Sol. **(B)**

Diameter

L : $x - 3y + 7 = 0$ ...(1)

Given points A(−8, 5) and B(6, 5) lies on same side of line L

Perpendicular bisector of AB will be another diameter

mid point of AB : M(−1, 5)

$m_{AB} = 0$

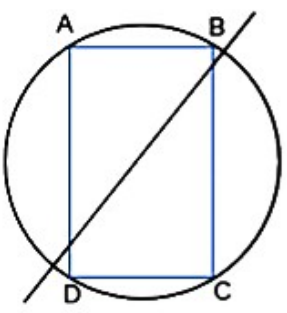
equation of perpendicular bisector AB : $x = -1$ ...(2)

On solving (1) and (2), centre O(−1, 2)

O is mid point of BD, hence D(−8, −1)

ar(ABCD) = AB × AD

= 14 × 6 = 84 sq. unit



Q **If line $y = x + c$ and $y^2 = 8x$ intersect in A and B. Circle with AB as diameter passes through (0, 0). Find c?**

Sol. By homogenization equation of pair of straight line

$$y^2 = 8x\left(\frac{y - x}{c}\right)$$

$$\Rightarrow 8x^2 + cy^2 - 8xy = 0$$

Lines are perpendicular ($\because$ circle with diameter passes through origin)

$\Rightarrow 8 + c = 0 \Rightarrow c = -8$

Q **Find locus of point of intersection of $x + 2y + \lambda(x - 2y) = 0$ and $(x + y - 2) + \lambda(x - 2) = 0$ if these lines are always perpendicular to each other.**

Sol. First family of straight lines $L_1 + \lambda L_2 = 0$ passes through point of intersection of

$L_1 : x + 2y = 0$ and $L_2 : x - 2y = 0$ which is A(0, 0)

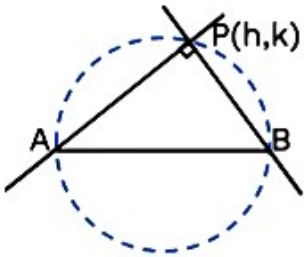
Second family of straight lines $L_3 + \lambda L_4 = 0$ passes through point of intersection of

$L_3 : x + y = 2$ and $L_4 : x - 2 = 0$ which is B(2, 0)

Now required locus will be circle as lines from A and B are perpendicular and A and B will be diametric end points

Equation : $(x - 0)(x - 2) + (y - 0)(y - 0) = 0$

$$x^2 + y^2 - 2x = 0$$

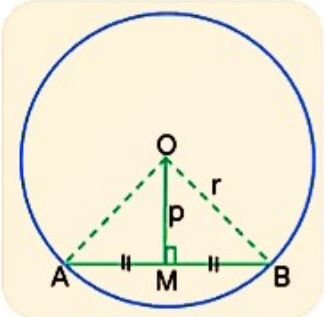


INTERCEPT

Length of chord:

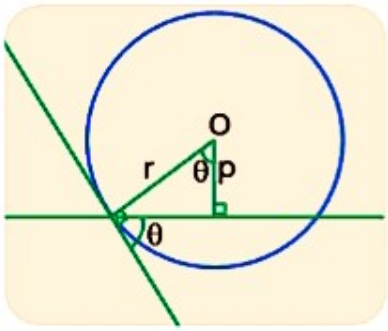
AB = 2MB

$$AB = 2\sqrt{r^2 - p^2}$$



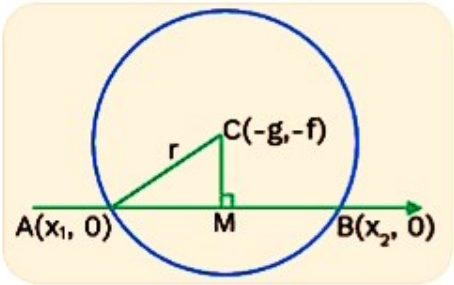
Angle between Line and circle:

$$\cos\theta = \frac{p}{r}$$



X-intercept:

$$|x_1 - x_2| = 2\sqrt{g^2 - c}$$



Proof :

$$AB = 2MB$$

$$= 2\sqrt{r^2 - p^2}$$

$$= 2\sqrt{\left(g^2 + f^2 - c\right) - \left(-f\right)^2}$$

$$= 2\sqrt{g^2 - c}$$

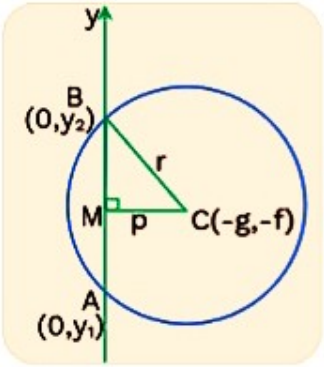
(i) If $g^2 > c$
$\Rightarrow$ Circle cuts the x-axis at 2 distinct points

(ii) If $g^2 = c$
$\Rightarrow$ Circle touches x-axis

(iii) If $g^2 < c$
$\Rightarrow$ Circle lies completely above or below the x-axis

Y-intercept :

$$|y_1 - y_2| = 2\sqrt{f^2 - c}$$



Proof :

$$AB = 2MB$$

$$= 2\sqrt{r^2 - p^2}$$

$$= 2\sqrt{\left(g^2 + f^2 - c\right) - \left(-g\right)^2}$$

$$= 2\sqrt{f^2 - c}$$

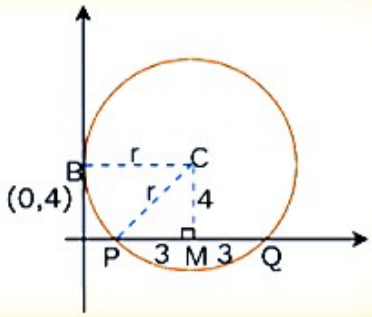
(i) If $f^2 > c$ $\Rightarrow$ circle cuts the y-axis at 2 distinct points

(ii) If $f^2 = c$ $\Rightarrow$ circle touches y-axis

(iii) If $f^2 < c$ $\Rightarrow$ circle lies completely either on right or on left of y-axis

Examples :

Q **Find the equation of the circle which touches the positive axis of y at a distance of 4 units from origin and cuts off an intercept of 6 unit from the positive axis.**



Sol. PQ = 6

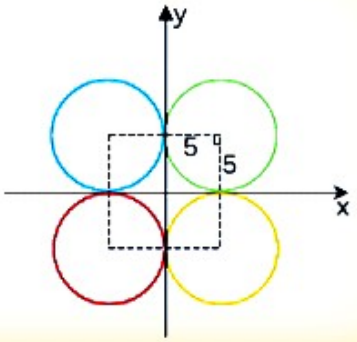
In ΔPCM

$r^2 = 3^2 + 4^2 \Rightarrow r = 5$

Centre : C(r, 4) ≡ C(5, 4)

Equation : $(x - 5)^2 + (y - 4)^2 = 5^2$

Q **Find the equation of circle which touches the co-ordinate axes and whose radius = 5.**



Sol. Equation:

$$(x \pm 5)^2 + (y \pm 5)^2 = 5^2$$

Q **Find the equation of a circle through origin cutting off intercept equals to unity on the lines $y^2 - x^2 = 0$.**

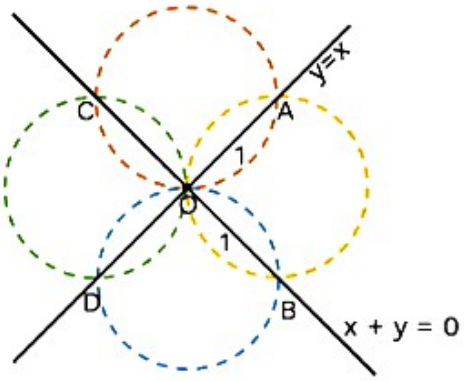
Sol. $y^2 - x^2 = (y + x)(y - x)$

$L_1 : x + y = 0$

$L_2 : x - y = 0$

$A\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), B\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

$C\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $D\left(\frac{-1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$



for required circles AB, AC, CD, BD are diameters

Equation: $\left(x \pm \frac{1}{\sqrt{2}}\right)\left(x \pm \frac{1}{\sqrt{2}}\right) + \left(y \pm \frac{1}{\sqrt{2}}\right)\left(y \pm \frac{1}{\sqrt{2}}\right) = 0$

$\Rightarrow x^2 + y^2 \pm \sqrt{2}x = 0$

or

$\Rightarrow x^2 + y^2 \pm \sqrt{2}y = 0$

Q **Find the equation of the locus of the centre of a circle which touches the positive y-axis and having intercept on x-axis equals to 2l.**

Sol. Let centre be P(h, k)

In ΔAPM

$r^2 = l^2 + k^2$ ...(1)

Also circle touches y-axis

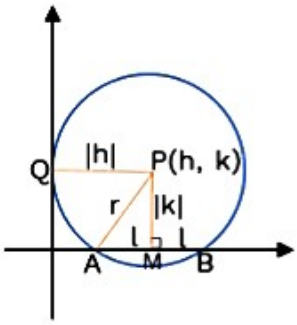
$\therefore$ QP = AP

$|h| = r \Rightarrow r^2 = h^2$ ...(2)

from (1) and (2),

$h^2 = l^2 + k^2 \Rightarrow h^2 - k^2 = l^2$

locus : $x^2 - y^2 = l^2$



Q **Find the equation of incircle and circumcircle of the quadrilateral formed by the lines x = 0 and y = 0, x = 16, y = 16.**

Sol. Centre of incircle S_1 and circumcircle S_2 will be mid-point of OA

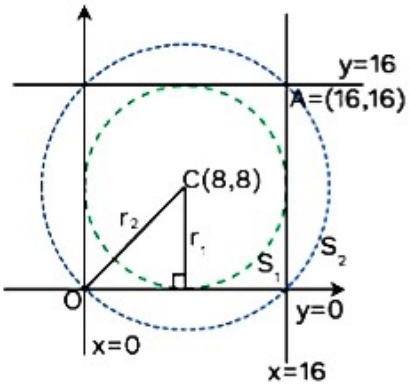
$\therefore$ centre C(8, 8)

For incircle, radius $r_1 = 8$

$S_1 : (x - 8)^2 + (y - 8)^2 = 8^2$

for circumcircle OA is diameter

$S_2 : (x - 0)(x - 16) + (y - 0)(y - 16) = 0$



Q **Find λ if length of intercept by line $3x - 4y + \lambda = 0$ on the circle $x^2 + y^2 = 25$ is of 8 unit.**

Sol. Centre O(0, 0), radius r = 5

$$p = \left|\frac{3(0) - 4(0) + \lambda}{\sqrt{3^2 + 4^2}}\right| \Rightarrow p = \frac{|\lambda|}{5}$$

length of intercept = $2\sqrt{r^2 - p^2}$

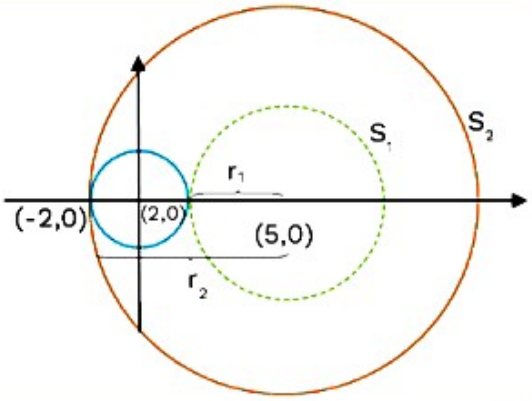
$$8 = 2\sqrt{25 - \frac{\lambda^2}{25}}$$

$$16 = 25 - \frac{\lambda^2}{25} \Rightarrow \frac{\lambda^2}{25} = 9 \Rightarrow \lambda = \pm 15$$

Q **Find the equation of circle whose centre is (5, 0) and touches the circle $x^2 + y^2 = 4$.**

Sol. Two circles are possible for S_1
centre (5, 0)
$r_1 = 3$
Equation: $(x - 5)^2 + y^2 = 3^2$

for S_2
centre (5, 0), $r_2 = 7$
Equation: $(x - 5)^2 + y^2 = 7^2$



Q **Two rods whose lengths are 2a and 2b move along the rectangular axes (one on X-axis and other on Y-axis) in such a way that their extremities are always concyclic. Find the equation of the locus of the centre of the circle.**

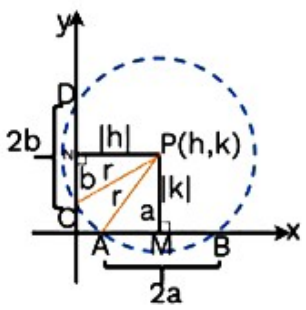
Sol. Let centre be P(h, k).
In ΔPMA,

$r^2 = a^2 + k^2$ ...(1)

In ΔPCN,

$r^2 = b^2 + h^2$ ...(2)

from (1) and (2),
$a^2 + k^2 = b^2 + h^2 \Rightarrow h^2 - k^2 = a^2 - b^2$
locus: $x^2 - y^2 = a^2 - b^2$



Q **If y = mx be the equation of a chord of a circle whose radius is 'a', the origin of co-ordinates being one extremity of the chord and the axis of x being a diameter of the circle. Prove that the equation of a circle of which this chord is the diameter is** $\left(1+m^2\right)\left(x^2+y^2\right)-2a\left(x+my\right)=0$

Sol. Equation of given circle

$x\left(x-2a\right)+y^2=0$

$x^2+y^2-2ax=0$ ...(1)

Chord, $y = mx$ ...(2)

from (1) and (2)

$x^2+m^2x^2-2ax=0$

$$x\left\{\left(1+m^2\right)x-2a\right\}=0 \Rightarrow x_1=\frac{2a}{1+m^2}$$

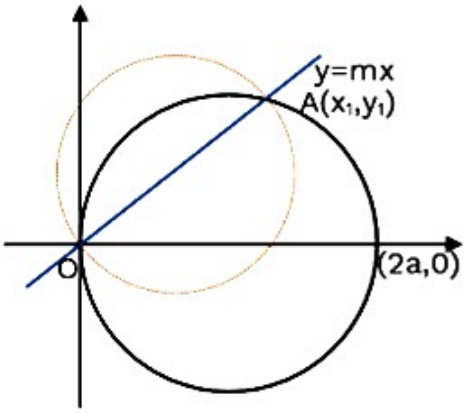
$$\because\ y_1=mx_1 \Rightarrow y_1=\frac{2am}{1+m^2}$$

for required circle OA is diameter hence

Equation: $x\left(x-x_1\right)+y\left(y-y_1\right)=0$

$$x^2+y^2-\frac{2ax}{1+m^2}-\frac{2amy}{1+m^2}=0$$

$$\left(1+m^2\right)\left(x^2+y^2\right)-2a\left(x+my\right)=0$$



Hence proved.

Q **Prove that the equation to the circle of which the points (x_1, y_1) and (x_2, y_2) are the ends of a chord of a segment containing an angle θ, is $(x_1 - x)(x - x_2) + (y - y_1)(y - y_2) \pm \cot\theta\,[(x - x_1)(y - y_2) - (x - x_2)(y - y_1)] = 0$**

Sol.

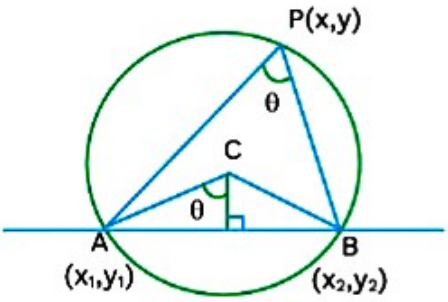
$$m_{AP}=\frac{y-y_1}{x-x_1}$$

$$m_{BP}=\frac{y-y_2}{x-x_2}$$

Now angle between AP and BP is θ, hence

$$\tan\theta=\pm\left(\frac{m_{AP}-m_{BP}}{1+m_{AP}m_{BP}}\right)$$

$$\pm\tan\theta=\frac{\frac{y-y_1}{x-x_1}-\frac{y-y_2}{x-x_2}}{1+\left(\frac{y-y_1}{x-x_1}\right)\left(\frac{y-y_2}{x-x_2}\right)}$$



$$\Rightarrow \pm\tan\theta = \frac{(y - y_1)(x - x_2) - (y - y_2)(x - x_1)}{(x - x_1)(x - x_2) + (y - y_1)(y - y_2)}$$

$$\Rightarrow (x - x_1)(x - x_2) + (y - y_1)(y - y_2) \pm \cot\theta\left[(x - x_1)(y - y_2) - (x - x_2)(y - y_1)\right] = 0$$

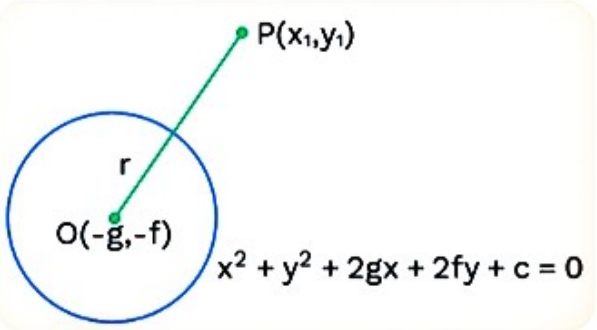
Hence proved.

Position of point w.r.t. a circle

$S_1 \equiv$ Power of point

$S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

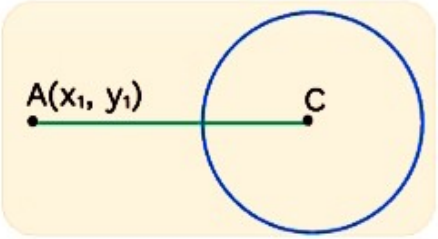
If $S_1 > 0 \Rightarrow$ Point exterior of circle
If $S_1 = 0 \Rightarrow$ Point on circle
If $S_1 < 0 \Rightarrow$ Point interior of circle.



Greatest and Least distance of a point $A(x_1, y_1)$

Maximum distance $= |AC + r|$

and Minimum distance $= |AC - r|$

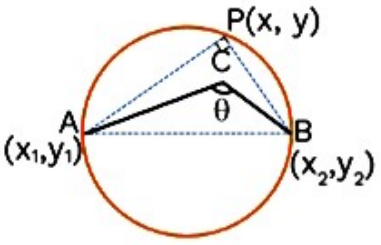


Example :

Q **If the join of (x_1, y_1) and (x_2, y_2) makes an obtuse angle at (x_3, y_3) then prove that $(x_3 - x_1)(x_3 - x_2) + (y_3 - y_1)(y_3 - y_2) < 0$.**

Sol. If AB is diameter of circle then its equation
$S : (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
Clearly AB makes obtuse angle at $C(x_3, y_3)$ only if C lies interior of the circle hence $S_1 < 0$.

$$(x_3 - x_1)(x_3 - x_2) + (y_3 - y_1)(y_3 - y_2) < 0$$



Q $S_1 = x^2 + y^2 - 4x + 6y - 3 = 0$

$S_2 = x^2 + y^2 + 4x - 6y - 3 = 0$

Point (1, 2) lies

(A) Inside $S_1 = 0$ and inside $S_2 = 0$ **(B) Outside $S_1 = 0$ and Outside $S_2 = 0$**

(C) Inside $S_1 = 0$ and Outside $S_2 = 0$ **(D) Outside $S_1 = 0$ and inside $S_2 = 0$**

Sol. **(D)**

for S_1

$1^2 + 2^2 - 4(1) + 6(2) - 3 > 0 \Rightarrow$ Outside

for S_2

$1^2 + 2^2 + 4(1) - 6(2) - 3 < 0 \Rightarrow$ Inside

Q **Find the minimum and maximum distance between two points one lying on the circle $x^2+y^2=144$ and other lying on $(x - 15)^2 + (y - 20)^2 = 1$**

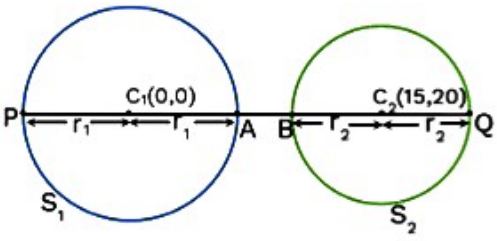
Sol. $C_1(0, 0)$, $r_1 = 12$

$C_2(15, 20)$, $r_2 = 1$

Also $C_1C_2 = \sqrt{15^2 + 20^2} = 25$

Minimum distance = AB = $C_1C_2 - r_1 - r_2$ = 12 unit

Maximum distance = PQ = $C_1C_2 + r_1 + r_2$ = 38 unit



Q4 **Find minimum and maximum distance between any point on circle $x^2 + y^2 = 25$ and point (6, 8)**

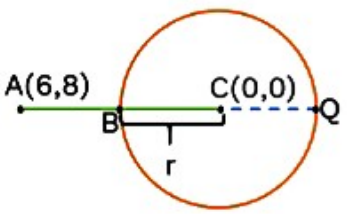
Sol. Minimum distance

AB = AC − r

$\Rightarrow$ AB = $\sqrt{6^2 + 8^2} - 5 = 5$ unit

Maximum distance AQ = AC + r

= 15 unit



Q **If the variable line $3x - 4y + k = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, then the range of k is (a, b) where a, b$\in$ I. Find the value of (b – a).**

Sol. For first circle

$C_1(1, 1)$, $r_1 = 1$

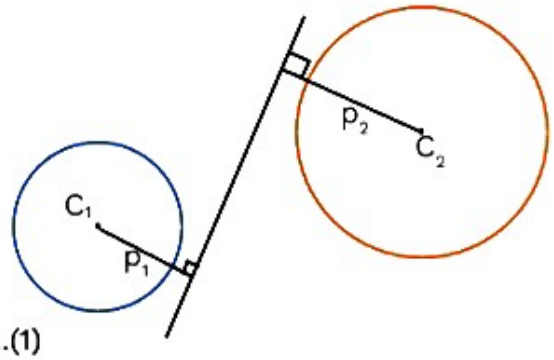
for second circle

$C_2(8, 1)$, $r_2 = 2$

Clearly $p_1 > r_1$

$$\left|\frac{3 - 4 + k}{5}\right| > 1 \Rightarrow |k - 1| > 5$$

$$k \in (-\infty, -4) \cup (6, \infty) \quad \ldots(1)$$



$p_2 > r_2$

$$\left|\frac{24 - 4 + k}{5}\right| > 2 \Rightarrow |20 + k| > 10$$

$$k \in (-\infty, -30) \cup (-10, \infty) \quad \ldots(2)$$

also C_1 and C_2 lies opposite side of line

$(3 - 4 + k)(24 - 4 + k) < 0$

$(k - 1)(k + 20) < 0 \Rightarrow k \in (-20, 1)$ ...(3)

(1) $\cap$ (2) $\cap$ (3)

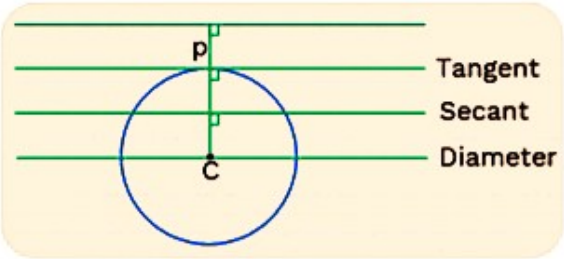
$\Rightarrow$ $k \in (-10, -4)$

hence $a = -10$, $b = -4$

$b - a = 10 - 4 = 6$

Line and a Circle

Let L = 0 be a line and S = 0 be a circle. If 'r' is the radius of the circle and 'p' is the length of perpendicular from the centre on the line, then



(i) If $p > r \Rightarrow$ line is neither secant nor tangent and passes outside the circle.

(ii) If $p = r \Rightarrow$ line is tangent to the circle.

(iii) If $p < r \Rightarrow$ line is secant.

(iv) If $p = 0 \Rightarrow$ line is a diameter.

II Method

Solve the line with the circle and if

(i) $D > 0 \Rightarrow$ line is Secant.

(ii) $D = 0 \Rightarrow$ line is Tangent.

(iii) $D < 0 \Rightarrow$ line passes outside the circle.

Example:

Q **For what value of 'm' the line $3x - my + 6 = 0$ is tangent to the circle $x^2 + y^2 - 4x + 6y - 3 = 0$.**

Sol. Centre C(2, –3), radius = $r = \sqrt{4 + 9 + 3}$

$r = 4$

Now $p = r$

$$\Rightarrow \left|\frac{3(2) - m(-3) + 6}{\sqrt{9 + m^2}}\right| = 4$$

$$\Rightarrow |12 + 3m| = 4\sqrt{9 + m^2}$$

On squaring,

$144 + 9m^2 + 72m = 144 + 16m^2$

$\Rightarrow 7m^2 - 72m = 0$

$\Rightarrow m = 0$ or $m = \frac{72}{7}$

Q **The sum of square of lengths of the chords intercepted on the circle $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in N$ where N is the set of all natural numbers, is**
(A) 320 **(B) 160** **(C) 105** **(D) 210**

Sol. **(D)**

Centre C(0, 0), radius r = 4

$$p = \left|\frac{0 + 0 - n}{\sqrt{2}}\right| \Rightarrow p = \frac{n}{\sqrt{2}}$$

Length of chord $= 2\sqrt{r^2 - p^2}$

$$= 2\sqrt{16 - \frac{n^2}{2}} = \sqrt{64 - 2n^2}$$

where n can be 1, 2, 3, 4, 5

sum of squares of length of chords

$$= \sum_{r=1}^{5}\left(64 - 2n^2\right)$$

$$= 64 \times 5 - 2\left(\frac{5 \times 6 \times 11}{6}\right) = 42 \times 5 = 210$$

Q **Find k if line $3x + 4y = k$ touch the circle $x^2 + y^2 - 10x = 0$.**

Sol. centre C(5, 0), radius r = 5

p = r

$$\left|\frac{3(5) + 4(0) - k}{5}\right| = 5 \Rightarrow |15 - k| = 25$$

$\Rightarrow k = -10, 40$

Q **Find the equation of T = 0 to circle $x^2 + y^2 = 4$ and parallel to line $x + 2y + 3 = 0$**

Sol. Let tangent be $x + 2y + k = 0$

centre C(0, 0), radius r = 2

p = r

$$\left|\frac{0 + 0 + k}{\sqrt{5}}\right| = 2 \Rightarrow k = \pm 2\sqrt{5}$$

equation of tangent: $x + 2y \pm 2\sqrt{5} = 0$

Q **Find radius of circle whose tangents are**
$6x + 8y + 26 = 0$
$3x + 4y - 17 = 0$

Sol. Tangents are $L_1 : 3x + 4y + 13 = 0$

$L_2 : 3x + 4y - 17 = 0$

Now diameter = distance between L_1 and L_2

$$2r = \left|\frac{13 + 17}{\sqrt{9 + 16}}\right|$$

$$2r = \frac{30}{5} \Rightarrow r = 3$$

Note :

(i) Number of tangents from external point to circle is 2.

(ii) Number of tangents from interior point to circle is 0.

(iii) If point is on periphery then number of tangent is 1.

Parametric Equation of a circle

$x = x_1 + r\cos\theta$ and $y = y_1 + r\sin\theta$

$(x_1, y_1) \rightarrow$ fixed centre,
$r \rightarrow$ fixed radius and
$\theta \in [0, 2\pi)$ is a parameter.

Note :

If θ is eliminated we get cartesian form of a circle i.e. $(x - x_1)^2 + (y - y_1)^2 = r^2$

Example :

Q **Convert $x^2 + y^2 - 6x + 4y - 3 = 0$ in parametric form.**

Sol. Centre $(3, -2) \equiv (x_1, y_1)$

$$r = \sqrt{9 + 4 + 3} \Rightarrow r = 4$$

parametric form

$x = x_1 + r\cos\theta, \quad y = y_1 + r\sin\theta$

$x = 3 + 4\cos\theta, \quad y = -2 + 4\sin\theta$

Q **If $x^2 + y^2 - 2x - 4y - 4 = 0$, find maximum and minimum value of $3x + 4y$.**

Sol. Centre C(1, 2), radius r = 3
Parametric form: $x = 1 + 3\cos\theta$, $y = 2 + 3\sin\theta$
$3x + 4y = 11 + (9\cos\theta + 12\sin\theta)$

$$11 - \sqrt{9^2 + 12^2} \le 3x + 4y \le 11 + \sqrt{9^2 + 12^2}$$

Min. value = 11 – 15 = – 4
Max. value = 11 + 15 = 26

Q **Find circumcentre of ΔABC where co-ordinates of**

$A \equiv \left(2+\cos\frac{\pi}{3}, 3+\sin\frac{\pi}{3}\right)$

$B \equiv (2+\cos\pi, 3+\sin\pi)$

$C \equiv \left(2+\cos\frac{4\pi}{3}, 3+\sin\frac{4\pi}{3}\right)$

Sol. Clearly all three vertices are of the form $(x_1 + r\cos\theta, y_1 + r\sin\theta)$

which lies on circle with centre
$(x_1, y_1) \equiv (2, 3)$ which is circumcentre of ΔABC.

Q **If $A(\cos\theta_1, \sin\theta_1)$; $B(\cos\theta_2, \sin\theta_2)$; $C(\cos\theta_3, \sin\theta_3)$ are the vertices of ΔABC then, find**
(i) Centroid of ΔABC

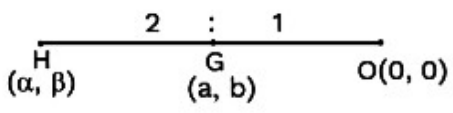
Sol. Centroid : $G\left(\frac{\cos\theta_1+\cos\theta_2+\cos\theta_3}{3}, \frac{\sin\theta_1+\sin\theta_2+\sin\theta_3}{3}\right)$

(ii) Circumcentre of ΔABC

Sol. A, B, C lies on circle $x^2 + y^2 = 1$
hence circumcentre is O(0, 0)

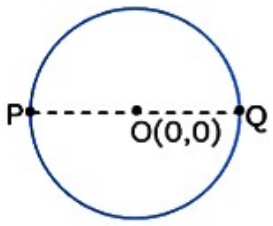
(iii) Orthocentre of ΔABC

Sol. Let orthocentre be $H(\alpha, \beta)$
$\alpha = 3a$, $\beta = 3b$
hence orthocentre
$H(\cos\theta_1 + \cos\theta_2 + \cos\theta_3 , \sin\theta_1 + \sin\theta_2 + \sin\theta_3)$



Q **Let PQ be diameter of the circle $x^2 + y^2 = 9$. If α and β are the lengths of the perpendiculars from P and Q on the straight line $x + y = 2$ respectively, then the maximum value of $\alpha\beta$ is**

Sol. Let P$(3\cos\theta, 3\sin\theta)$
also PQ is diameter and O(0, 0) is mid-point of PQ hence
Q$(-3\cos\theta, -3\sin\theta)$



$$\text{Now } \alpha = \left|\frac{3\cos\theta + 3\sin\theta - 2}{\sqrt{2}}\right| \quad \ldots(1)$$

$$\beta = \left|\frac{-3\cos\theta - 3\sin\theta - 2}{\sqrt{2}}\right| \quad \ldots(2)$$

$$\alpha\beta = \left|\frac{(3\cos\theta + 3\sin\theta)^2 - 4}{2}\right| = \left|\frac{9(1+\sin 2\theta) - 4}{2}\right|$$

$$\alpha\beta = \left|\frac{5 + 9\sin 2\theta}{2}\right|$$

$$\frac{5-9}{2} \le \frac{5+9\sin 2\theta}{2} \le \frac{5+9}{2}$$

$$-2 \le \frac{5+9\sin 2\theta}{2} \le 7 \Rightarrow 0 \le \left|\frac{5+9\sin 2\theta}{2}\right| \le 7$$

$$\alpha\beta_{min} = 0, \alpha\beta_{max} = 7$$

Q **Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value (2M + 6m)**

Sol. Point on circle
Let $x = \cos\theta$, $y = \sin\theta$

$$\text{Now } z = \frac{4 - \sin\theta}{7 - \cos\theta}$$

$$z = \frac{4 - \dfrac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}}{7 - \dfrac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}} = \frac{4t^2 - 2t + 4}{8t^2 + 6} \qquad \left(t = \frac{\tan\theta}{2} \in R\right)$$

$$z = \frac{2t^2 - t + 2}{4t^2 + 3}$$

$$(4z-2)t^2 + t + (3z-2) = 0$$

$D \geq 0$

$1 - 4(4z - 2)(3z - 2) \geq 0$

$1 - (48z^2 - 56z + 16) \geq 0$

$48z^2 - 56z + 15 \leq 0$

$48z^2 - 36z - 20z + 15 \leq 0$

$12z(4z - 3) - 5(4z - 3) \leq 0$

$(12z - 5)(4z - 3) \leq 0$

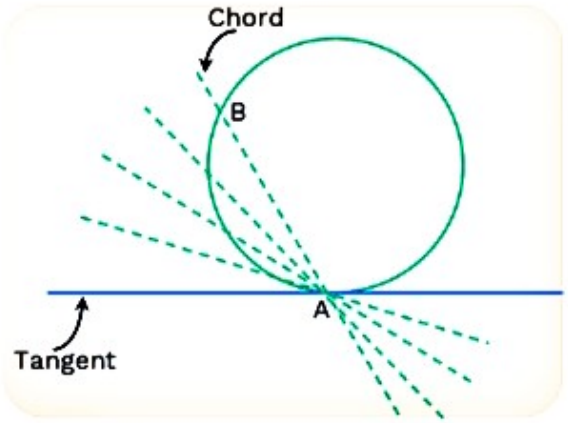
$$\Rightarrow z \in \left[\frac{5}{12}, \frac{3}{4}\right]$$

$$M = \frac{3}{4},\ m = \frac{5}{12}$$

$$2M + 6m = \frac{3}{2} + \frac{5}{2} = 4$$

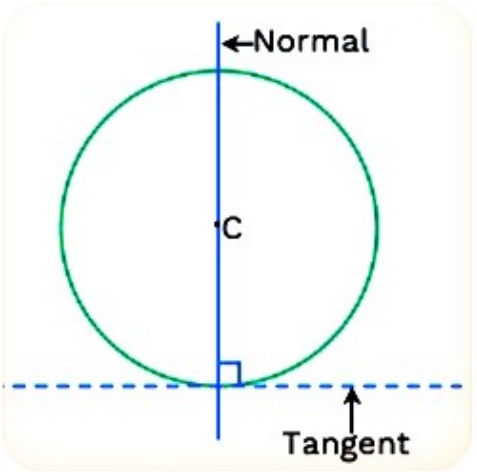
Tangent :

Tangent is the limiting case of the secant as the point B tends to A.



Normal :

Normal is the line perpendicular to the tangent passing through the point of tangency. In case of circle, normal always passes through centre.

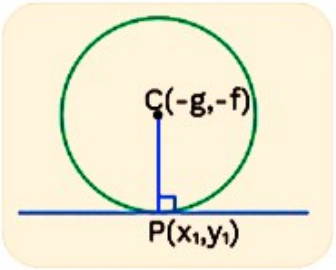


Equation of the tangent drawn to the circle in the Various forms :

Cartesian Form :

Tangent drawn to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at its point (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Proof :



Centre of circle C(–g, –f)

Tangent at point $P(x_1, y_1)$ will be perpendicular to radius CP, hence

$$m_{cp} = \frac{y_1 + f}{x_1 + g}$$

Slope of tangent at P

$$m_T = -\left(\frac{x_1 + g}{y_1 + f}\right)$$

Equation: $(y - y_1) = -\left(\frac{x_1 + g}{y_1 + f}\right)(x - x_1)$

$$(y - y_1)(y_1 + f) + (x_1 + g)(x - x_1) = 0$$

$$yy_1 + fy - y_1^2 - fy_1 + xx_1 - x_1^2 + gx - gx_1 = 0$$

$$\Rightarrow xx_1 + yy_1 + gx + fy = x_1^2 + y_1^2 + gx_1 + fy_1 \quad \ldots(1)$$

Also P lies on circle hence

$$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$$

$$x_1^2 + y_1^2 + gx_1 + fy_1 = -gx_1 - fy_1 - c \quad \ldots(2)$$

from (2) into (1)

Equation: $xx_1 + yy_1 + gx + fy = -gx_1 - fy_1 - c$

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

Hence proved.

Cartesian form of circle with centre origin

If circle is $x^2 + y^2 = a^2$ then equation of tangent is $xx_1 + yy_1 = a^2$.

Example :

Q **If equation of circle $x^2 + y^2 = 25$. Find equation of tangent at (3, 4)**

Sol. Equation is

$$x(3) + y(4) = 25 \Rightarrow 3x + 4y = 25$$

Parametric form :

$$\left.\begin{aligned} x_1 &= r\cos\theta \\ y_1 &= r\sin\theta \end{aligned}\right\}, \quad [0, 2\pi)$$

Equation of tangent is

$x\cos\theta + y\sin\theta = r$

Proof :

Equation using Cartesian form

$x(r\cos\theta) + y(r\sin\theta) = r^2$

$\Rightarrow \quad x\cos\theta + y\sin\theta = r$

Slope Form :

$x^2 + y^2 = a^2$

Equation of the tangent is

$y = mx \pm a\sqrt{1+m^2}$

Proof :

Let equation of tangent is $y = mx + c$

Now $p = r$

$$\left|\frac{m(0) - 0 + c}{\sqrt{m^2 + 1}}\right| = a \Rightarrow c = \pm a\sqrt{1+m^2}$$

Equation: $y = mx \pm a\sqrt{1+m^2}$ **Hence proved.**

Note :

For a unique value of m there will be 2 tangent which are parallel to each other.

Point of Tangency

Method-I

Step-1:

Write equation of normal {Perpendicular to T = 0 and passing through (-g, -f)}

Step-2:

Intersection of N = 0, T = 0 is co-ordinate of that point.

Method-II

Compare with point form.

Example :

Q **Find point of tangency if equation of tangent is $3x + 4y = 50$ to the circle $x^2 + y^2 - 6x - 8y = 0$**

Sol. Equation of Normal is

$4x - 3y = \lambda$

It passes through centre (3, 4) hence

$4(3) - 3(4) = \lambda \Rightarrow \lambda = 0$

Normal : $4x = 3y$

Tangent : $3x + 4y = 50$

On solving, $3x + 4\left(\frac{4x}{3}\right) = 50 \Rightarrow 25x = 150$

$x = 6, y = 8$

Point of tangency : (6, 8)

Q **A circle passes through the points (-1, 1), (0, 6) and (5, 5). Find the points on this circle, the tangent at which are parallel to the straight line joining the origin to its centre.**

Sol. Let A(–1, 1), B(0, 6) and C(5, 5)

$AB = \sqrt{5^2 + 1^2} = \sqrt{26}$

$BC = \sqrt{5^2 + 1^2} = \sqrt{26}$

$AC = \sqrt{6^2 + 4^2} = \sqrt{52}$

$\because AC^2 = AB^2 + BC^2 \Rightarrow \Delta ABC$ is right angled

$\therefore$ Circumcentre (mid-point of AC) is P(2, 3)

radius = $r = PA = \sqrt{3^2 + 2^2} = \sqrt{13}$

equation : $(x-2)^2 + (y-3)^2 = 13$ ...(1)

$x^2 + y^2 - 4x - 6y = 0$

Slope of line joining O and P: $m_{op} = \frac{3}{2}$

equation of normal: $(y-3) = \frac{-2}{3}(x-2)$ ...(2)

by (1) and (2), $(x-2)^2 + \frac{4}{9}(x-2)^2 = 13$

$(x-2)^2 = 9 \Rightarrow x = 5, -1$

$y = 1, 5$

required points are (5, 1) and (–1, 5)

Q **Find the equation of the tangents to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ which are**
(i) Perpendicular to the line $3x - 4y + 7 = 0$.

Sol. Centre (1, 2), radius $r = \sqrt{1+4+4} = 3$
Let tangent is $4x + 3y + k = 0$
Now $p = r$

$$\left|\frac{4(1) + 3(2) + k}{5}\right| = 3 \Rightarrow |10 + k| = 15$$

$K = 5, -25$
Equation: $4x + 3y + 5 = 0$, $4x + 3y - 25 = 0$.

(ii) Parallel to the line $3x - 4y + 7 = 0$.

Sol. Let tangent be $3x - 4y + k = 0$

now $p = r$

$$\left|\frac{3(1) - 4(2) + k}{5}\right| = 3$$

$$|k - 5| = 15 \Rightarrow k = 20, -10$$

equation: $3x - 4y + 20 = 0$, $3x - 4y - 10 = 0$

Q **Find the equation of the tangent to the circle $x^2 + y^2 = 4$ drawn from the point (2, 3).**

Sol. Centre O(0, 0), radius $r = 2$
Let tangent be $(y - 3) = m(x - 2)$
$mx - y + 3 - 2m = 0$
Now $P = r$

$$\left|\frac{m(0) - 0 + 3 - 2m}{\sqrt{m^2 + 1}}\right| = 2$$

$$|3 - 2m| = 2\sqrt{m^2 + 1}$$

on squaring,

$9 + 4m^2 - 12m = 4m^2 + 4$

$\Rightarrow m = \frac{5}{12}$

Clearly given point (2, 3) is outside the circle hence two tangents can be drawn in which one tangent have slope $\frac{5}{12}$ and other must be parallel to y-axis

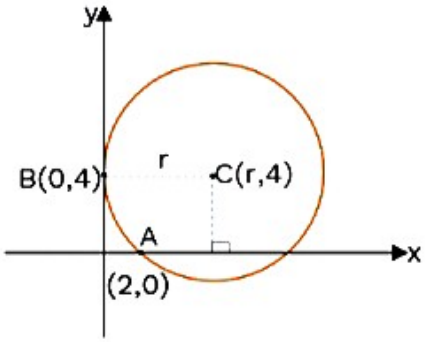
$\therefore$ equation : x= 2, $(y-3) = \frac{5}{12}(x-2)$

Q **A circle touches the y-axis at the point (0, 4) and passes through the point (2, 0). Which of the following lines is not a tangent to this circle?**

(A) 4x – 3y + 17 = 0 **(B) 3x + 4y – 6 = 0**

(C) 4x + 3y – 8 = 0 **(D) 3x – 4y – 24 = 0**

Sol. **(C)**



AC = r

$\Rightarrow AC^2 = r^2$

$(r-2)^2 + 4^2 = r^2$

$r^2 - 4r + 4 + 16 = r^2$

$4r = 20$

$r = 5$

equation of circle :

$(x-5)^2 + (y-4)^2 = 5^2$

centre C(5, 4), r = 5

Now check p = r for options

(A) $p = \left|\frac{20 - 12 + 17}{5}\right| = 5 = r$ ☑

(B) $p = \left|\frac{15 + 16 - 6}{5}\right| = 5 = r$ ☑

(C) $p = \left|\frac{20 + 12 - 8}{5}\right| = \frac{24}{5} \neq r$ $\because$

(D) $p = \left|\frac{15 - 16 - 24}{5}\right| = 5 = r$ ☑

Q **Find the equation of the tangent drawn to the circle $x^2 + y^2 - 6x + 4y - 3 = 0$ from the point (7, 4) lying outside the circle. Also find the point of contact.**

Sol. Centre (3, –2), radius $r = \sqrt{9 + 4 + 3} = 4$

Circle S : $(x - 3)^2 + (y + 2)^2 = 16$ ...(1)

equation of tangent from slope from

$$(y + 2) = m(x - 3) \pm r\sqrt{1 + m^2}$$

$$(y + 2) = m(x - 3) \pm 4\sqrt{1 + m^2}$$

Which passes through (7, 4) hence

$$6 = 4m \pm 4\sqrt{1 + m^2}$$

$$(3 - 2m)^2 = 4(1 + m^2)$$

$$9 - 12m + 4m^2 = 4 + 4m^2$$

$$\Rightarrow m = \frac{5}{12} \text{ and } m = \text{Not defined}$$

equation $(y + 2) = \frac{5}{12}(x - 3) \pm \frac{13}{3}$, $x = 7$

but (–) sign tangent does not-satisfy point (7, 4)

Hence, equation $T_1 : (y + 2) = \frac{5}{12}(x - 3) + \frac{13}{3}$ ...(2)

$T_2 : x = 7$ ...(3)

for point of contact

for $T_1 : 5x - 12y + 13 = 0$

$N_1 : 12x + 5y - 26 = 0$

on solving point of contact is $\left(\frac{95}{13}, \frac{22}{13}\right)$

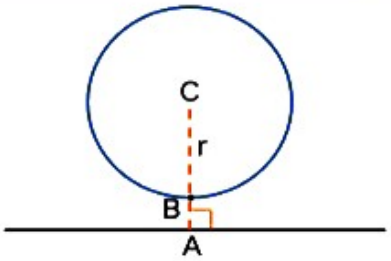
for $T_2 : x = 7$

put in equation of circle $4^2 + (y + 2)^2 = 4^2 \Rightarrow y = -2$

$\therefore$ point (7, –2)

Q **Find shortest distance between line $3x + 4y = 25$ and circle $x^2 + y^2 - 6x + 8y = 0$.**

Sol. Centre C(3, −4), radius $r = \sqrt{9+16} = 5$



$$p = \left|\frac{3(3) + 4(-4) - 25}{5}\right|$$

$$p = \frac{32}{5}$$

Shortest Distance = AB = p − r

$$= \frac{32}{5} - 5 = \frac{7}{5} \text{ unit}$$

Q **If equation of tangent line on circle $x^2 + y^2 = 1$ is $y = x + \sqrt{2}$ then find point of contact.**

Sol. Let point be $P(x_1, y_1)$ then tangent

$xx_1 + yy_1 = 1$ ...(1)

Also given equation of tangent

$x - y = -\sqrt{2}$ ...(2)

By comparing (1) and (2),

$$\frac{x_1}{1} = \frac{y_1}{-1} = \frac{1}{-\sqrt{2}}$$

$$\Rightarrow x_1 = -\frac{1}{\sqrt{2}}, y_1 = \frac{1}{\sqrt{2}}$$

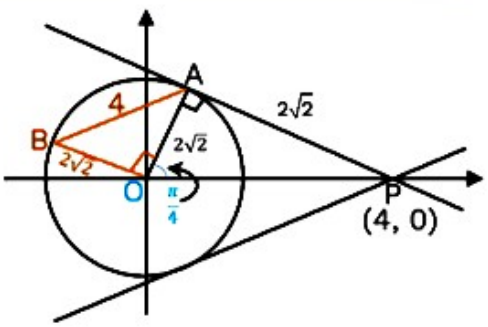
point is $\left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Q **Tangent is drawn from the point P(4, 0) to the circle $x^2 + y^2 = 8$ touches it at the point A in the 1st quadrant. Find the co-ordinates of another point B on the circle such that AB = 4.**

Sol. $OA = 2\sqrt{2}, OP = 4$

$\Rightarrow AP = 2\sqrt{2}$

hence $\angle AOP = \frac{\pi}{4}$



hence $A\left(2\sqrt{2}\cos\frac{\pi}{4}, 2\sqrt{2}\sin\frac{\pi}{4}\right) = A(2,2)$

By diagram we can see

$B\left(2\sqrt{2}\cos\frac{3\pi}{4}, 2\sqrt{2}\sin\frac{3\pi}{4}\right)$ or $B\left(-2\sqrt{2}\cos\frac{3\pi}{4}, -2\sqrt{2}\sin\frac{3\pi}{4}\right)$

$\Rightarrow$ B(−2, 2) or B(2, −2)

Q **Find the locus of the middle points of portions of tangents to the circle $x^2 + y^2 = a^2$ terminated by the co-ordinate axes.**

Sol. Let mid-point be P(h, k) and equation of tangent is

$x\cos\theta + y\sin\theta = a$ ...(1)

which meets x-axis at $A\left(\frac{a}{\cos\theta}, 0\right)$, $B\left(0, \frac{a}{\sin\theta}\right)$

now P is mid-point of AB hence

$\frac{a}{2\cos\theta} = h, \frac{b}{2\sin\theta} = k$

$\left(\frac{a}{2h}\right)^2 + \left(\frac{b}{2k}\right)^2 = 1 \Rightarrow \frac{a^2}{h^2} + \frac{b^2}{k^2} = 4$

Locus: $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$

Q **Let RS be the diameter of the circle $x^2 + y^2 = 1$ where S is the point (1, 0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. Then the locus of E passes through the point(s)**

(A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ **(B)** $\left(\frac{1}{4}, \frac{1}{2}\right)$ **(C)** $\left(\frac{1}{3}, \frac{-1}{\sqrt{3}}\right)$ **(D)** $\left(\frac{1}{4}, \frac{-1}{2}\right)$

Sol. **(AC)**

Let P $(\cos\theta, \sin\theta)$

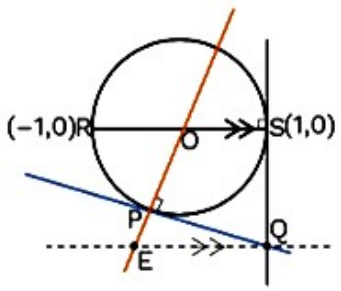
equation of tangent at P:

$x\cos\theta + y\sin\theta = 1$ ...(1)

equation of tangent at S(1, 0)

$x = 1$ ...(2)

from (1) and (2)



$$y = \frac{1 - \cos\theta}{\sin\theta}$$

$$= \frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$

hence $Q\left(1, \tan\frac{\theta}{2}\right)$

equation of line from Q parallel to RS is

$$y = \tan\frac{\theta}{2} \qquad ...(3)$$

Normal at P has equation

$$y = (\tan\theta)x \qquad ...(4)$$

from (3) and (4)

$$\tan\theta \cdot x = \tan\frac{\theta}{2} \Rightarrow x = \frac{1 - \tan^2\frac{\theta}{2}}{2}$$

$$\text{hence } E = \left(\frac{1 - \tan^2\frac{\theta}{2}}{2}, \tan\frac{\theta}{2}\right) \equiv (h,k)$$

$$h = \frac{1 - k^2}{2} \Rightarrow k^2 + 2h = 1$$

Locus : $y^2 + 2x = 1$

Note :

Point of intersection of the tangent drawn to the circle $x^2 + y^2 = a^2$ at the point

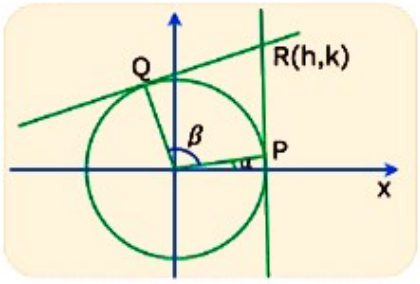
$P(\alpha)$ and $Q(\beta)$ is

$P(\alpha) \equiv (a\cos\alpha, a\sin\alpha)$

$Q(\beta) \equiv (a\cos\beta, a\sin\beta)$

$$h = \frac{a\cos\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha - \beta}{2}\right)}; \quad k = \frac{a\sin\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha - \beta}{2}\right)}$$

Proof :



tangent at P

T_1 : $x\cos\alpha + y\sin\alpha = a$ ...(1)

tangent at Q

T_2 : $x\cos\beta + y\sin\beta = a$ ...(2)

Multiply (1) by $\sin\beta$ and subtract (2) from it after multiplying by $\sin\alpha$,

$x\sin(\beta - \alpha) = a(\sin\beta - \sin\alpha)$

on simplifying we get

$$x = \frac{a\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}$$

Similarly, $y = \dfrac{a\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}$

Example :

Q **Find the locus of the point of intersection of the pair of tangents drawn to a circle $x^2 + y^2 = a^2$ at P(α) and Q(β) where $|\alpha - \beta| = 120°$.**

Sol. Clearly $h = \dfrac{a\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos(\pm 60°)}$

$$\Rightarrow a\cos\left(\frac{\alpha+\beta}{2}\right) = \frac{h}{2} \quad ...(1)$$

Also $k = \dfrac{a\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos(\pm 60°)}$

$$\Rightarrow a\sin\left(\frac{\alpha+\beta}{2}\right) = \frac{k}{2} \quad ...(2)$$

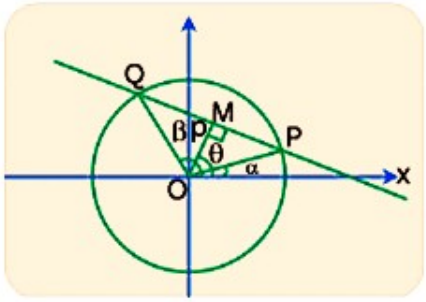
$(1)^2 + (2)^2 \Rightarrow a^2 = \dfrac{1}{4}(h^2 + k^2)$

locus : $x^2 + y^2 = 4a^2$

Note :

Equation of a chord line joining two points α and β on the $x^2 + y^2 = a^2$ is

$$x\cos\left(\frac{\alpha+\beta}{2}\right) + y\sin\left(\frac{\alpha+\beta}{2}\right) = a\cos\left(\frac{\alpha-\beta}{2}\right)$$



Proof :

Equation: $x\cos\theta + y\sin\theta = p$

where $\theta = \frac{\alpha+\beta}{2}$

and in ΔOMP,

$$\cos(\theta - \alpha) = \frac{p}{OP}$$

$$p = a\cos\left(\frac{\beta-\alpha}{2}\right) \text{ or } a\cos\left(\frac{\alpha-\beta}{2}\right)$$

Example :

Q **In a ΔABC the equation of line BC is $x - y = 0$, O(2, 3), H(5, 8). Find equation of circumcircle.**

Sol. Image of orthocentre in any side of Δ lies on the circumcircle of the Δ

Image of H(5, 8) in $x - y = 0$ is H'(8, 5)

which lies on circumcircle hence

radius $r = OH' = \sqrt{6^2 + 2^2} = \sqrt{40}$

equation $(x-2)^2 + (y-3)^2 = 40$

Length of Tangent and Power of a Point :

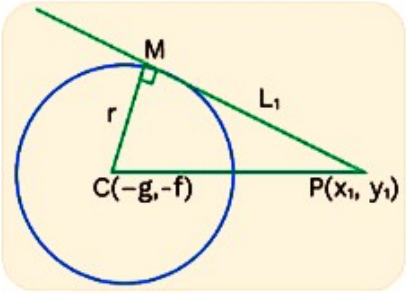
Length of the tangent from an external point (x_1, y_1) to a given circle.

$S = x^2 + y^2 + 2gx + 2fy + c = 0$ is given as

$$L_1 = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{S_1}$$

Power of point = Square of length of tangent = S_1

Proof :



$$L_1 = \sqrt{CP^2 - r^2}$$

$$L_1 = \sqrt{(x_1 + g)^2 + (y_1 + f)^2 - (g^2 + f^2 - C)}$$

$$L_1 = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{S_1}$$ **Hence proved.**

Example :

Q **Find length of tangent from (6, 8) to circle $x^2 + y^2 = 25$**

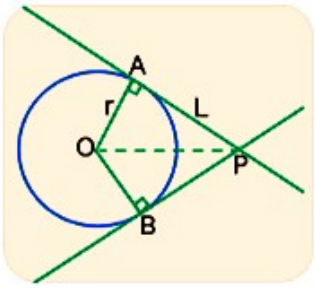
Sol. Length of $T = \sqrt{S_1}$

$= \sqrt{6^2 + 8^2 - 25} = 5\sqrt{3}$ unit

5-Important Deduction

(i) Area of Quad PAOB = rL

Proof :



Area of Quad PAOB

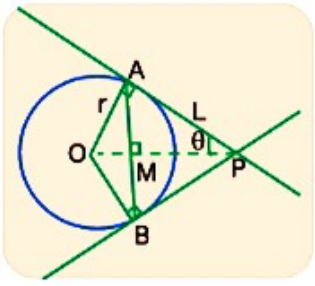
$= ar(\Delta PAO) + ar(\Delta PBO)$

$= \frac{1}{2}rL + \frac{1}{2}rL$

$= rL$

(ii) Area of $\Delta PAB = \frac{rL^3}{r^2 + L^2}$

Proof :



$OP = \sqrt{r^2 + L^2}$

$\sin\theta = \frac{r}{\sqrt{r^2 + L^2}}$

$\cos\theta = \frac{L}{\sqrt{r^2 + L^2}}$

$ar(\Delta PAB) = \frac{1}{2}AB \cdot MP$

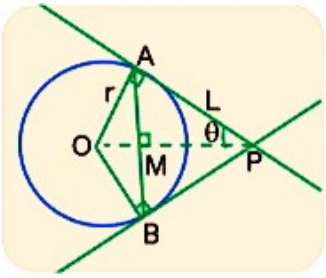
$= \frac{1}{2}(2AM)(L\cos\theta)$

$= (L\sin\theta)\,.\,(L\cos\theta)$

$$= \frac{rL^3}{r^2 + L^2}$$

(iii) Length of chord of contact

$$AB = \frac{2rL}{\sqrt{r^2 + L^2}}$$



Proof :

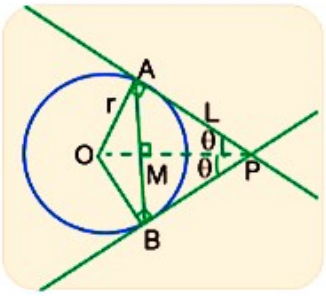
$AB = 2AM$

$= 2L\sin\theta$

$$= 2L\left(\frac{r}{\sqrt{r^2 + L^2}}\right)$$

(iv) Angle 2θ between the pair of tangent

$$2\theta = \tan^{-1}\left(\frac{2rL}{L^2 - r^2}\right)$$



Proof :

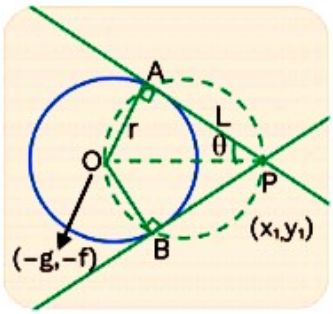
$$\tan\theta = \frac{r}{L}$$

$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

$$= \frac{2\left(\frac{r}{L}\right)}{1 - \left(\frac{r}{L}\right)^2} = \frac{2rL}{L^2 - r^2}$$

(v) Equation of the circle circumscribing the ΔPAB is

$$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$$



Proof :

For circumcircle of ΔPAB

$\because \angle OAP = 90°$

Hence OP will be diameter of the circle

Now equation is

$$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$$

Example :

Q **Tangents PA and PB are drawn from P(4, 3) to circle $x^2 + y^2 = 3^2$. Find**

(a) PA

Sol. $PA = \sqrt{S_1}$

$= \sqrt{4^2 + 3^2 - 3^2} = 4$

$\Rightarrow L = PA = 4$

(b) Area of quadrilateral PAOB

Sol. Area = rL

$= 3 \times 4 = 12$ sq. units.

(c) AB

Sol. $AB = \dfrac{2rL}{\sqrt{r^2 + L^2}} = \dfrac{2(3)(4)}{\sqrt{3^2 + 4^2}}$

$= \dfrac{24}{5}$ sq. unit.

(d) area of ΔPAB

Sol. $ar(\Delta PAB) = \dfrac{rL^3}{r^2 + L^2} = \dfrac{3(4)^3}{9 + 16} = \dfrac{192}{25}$ sq. unit

(e) ∠APB

Sol. $2\theta = \tan^{-1}\left(\dfrac{2rL}{L^2 - r^2}\right)$

$= \tan^{-1}\left(\dfrac{2(3)(4)}{(16 - 9)}\right) = \tan^{-1}\left(\dfrac{24}{7}\right)$

(f) equation of circumcircle of ΔPAB

Sol. endpoints of diameter are P(4, 3), O(0,0)

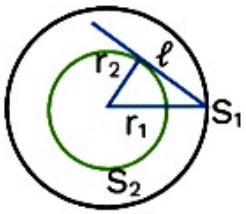
Equation: $(x-4)(x-0) + (y-3)(y-0) = 0$

$x^2 + y^2 - 4x - 3y = 0$

Q **Find the length of the tangent from any point on the circle $x^2 + y^2 = 25$ to the circle $x^2 + y^2 = 16$.**

Sol. Let point on first circle be $P(x_1, y_1)$

hence $x_1^2 + y_1^2 = 25$ ...(1)



Length of tangent $= \sqrt{S_2}$

$= \sqrt{x_1^2 + y_1^2 - 16}$

$= \sqrt{25 - 16} = 3$ unit

Alternative method : $\ell = \sqrt{r_1^2 - r_2^2}$

$\ell = \sqrt{25 - 16} = 3$ unit

Q **Find the range of 'p' for which the power of a point P(2, 5) is negative w.r.t. a circle $x^2 + y^2 - 8x - 12y + p = 0$ and the circle neither touches nor intersects the co-ordinate axis.**

Sol. Centre C(4, 6), radius $r = \sqrt{16 + 36 - p}$

$r = \sqrt{52 - p}$

$\because$ P point lies inside the circle

$\because S_1 < 0$

$2^2 + 5^2 - 16 - 60 + p < 0$

$\Rightarrow p < 47$ ...(1)

also circle neither intersects nor touches co-ordinate axis hence

$r < 4$

$\sqrt{52 - p} < 4 \Rightarrow 52 - p < 16$

$p > 36$ ...(2)

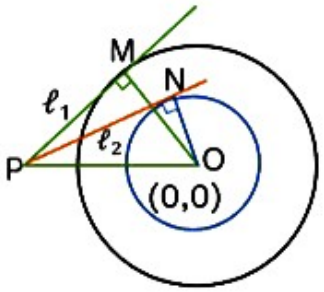
by (1) $\cap$ (2),

$36 < p < 47$

Q **Find the locus of a point, the tangents from which to the circle $4x^2 + 4y^2 - 9 = 0$ and $9x^2 + 9y^2 - 16 = 0$ are in the ratio 3 : 4.**

Sol. Let the point be P(h, k)

Equations of circles:



$$S_1 : x^2 + y^2 = \frac{9}{4}$$

$$S_2 : x^2 + y^2 = \frac{16}{9}$$

$$\frac{\ell_1}{\ell_2} = \frac{3}{4} \Rightarrow \ell_1 = 3\alpha, \ell_2 = 4\alpha$$

In ΔPMO,

$$\ell_1^2 = OP^2 - r_1^2 = OP^2 - \frac{9}{4} \quad \text{...(1)}$$

In ΔPNO,

$$\ell_2^2 = OP^2 - r_2^2 = OP^2 - \frac{16}{9} \quad \text{...(2)}$$

Dividing (1) by (2),

$$\frac{OP^2 - \frac{9}{4}}{OP^2 - \frac{16}{9}} = \frac{9}{16}$$

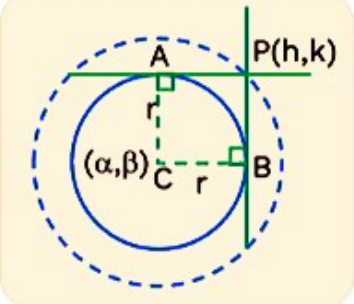
$$16OP^2 - 36 = 9OP^2 - 16$$

$$7OP^2 = 20 \Rightarrow h^2 + k^2 = \frac{20}{7}$$

locus: $x^2 + y^2 = \frac{20}{7}$

Director Circle :

Locus of intersection of two mutually perpendicular tangents.



$$(x - \alpha)^2 + (y - \beta)^2 = 2r^2$$

Proof :

PACB is a square

$PC = r\sqrt{2}$

$$(h - \alpha)^2 + (k - \beta)^2 = (r\sqrt{2})^2$$

locus : $(x - \alpha)^2 + (y - \beta)^2 = 2r^2$

Example :

Q **Find the range of 'a' such that the angle 'θ' between the pair of tangents drawn from the point (a, 0) to the circle $x^2 + y^2 = 4$ satisfies $\frac{\pi}{2} < \theta < \pi$**

Sol. for $\theta = \frac{\pi}{2}$, P lies on director circle

$\Rightarrow$ OP = $2\sqrt{2}$

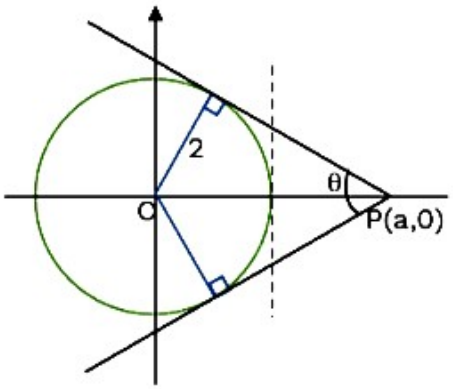
also for $\theta = \pi$

OP = 2

hence $2 < OP < 2\sqrt{2}$

$2 < |a| < 2\sqrt{2}$

$\Rightarrow a \in \left(-2\sqrt{2}, -2\right) \cup \left(2, 2\sqrt{2}\right)$



Chord in terms of Mid-Point

Let any chord PQ of the circle

$x^2 + y^2 + 2gx + 2fy + c = 0$ be bisected at $N(x_1, y_1)$.

Then slope of $MN = \frac{y_1 + f}{x_1 + g}$

hence, slope of the chord PQ is $-\frac{x_1 + g}{y_1 + f}$

then equation of PQ is $y - y_1 = -\frac{x_1 + g}{y_1 + f}(x - x_1)$

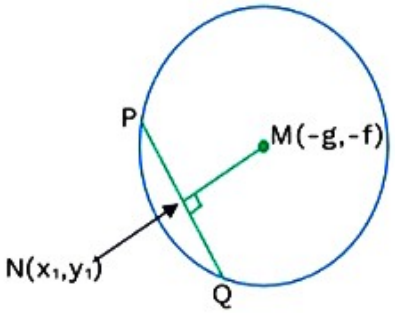
$\Rightarrow$ $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$

$= x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

$\Rightarrow$ $T = S_1$

where $T = xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$ and

$S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$



Examples :

Q **Find mid points of chord $2x - 5y + 18 = 0$ of the circle $x^2 + y^2 - 6x + 2y - 54 = 0$**

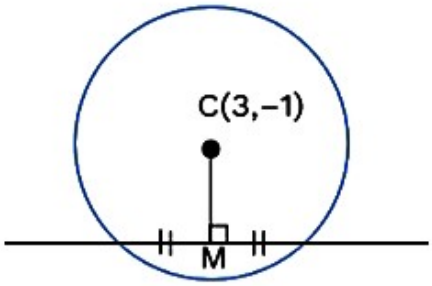
Sol. Centre C(3, –1)

Midpoint will be foot of perpendicular from centre on chord hence,

$$\frac{x-3}{2} = \frac{y+1}{-5} = -1\left\{\frac{2(3) - 5(-1) + 18}{4 + 25}\right\}$$

$$\frac{x-3}{2} = \frac{y+1}{-5} = -1 \Rightarrow x = 1,\ y = 4$$

midpoint M(1, 4)



Q **Locus of the middle point of the chords of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ which passes through a fixed point (a, b) lying outside the circle.**

Sol. Let mid point be M(h, k)

equation of chord : $T = S_1$

$$xh + yk + g(x + h) + f(y + k) + c = h^2 + k^2 + 2gh + 2fk + c$$

Also this chord passes through (a, b) hence

$ah + bk + g(a + h) + f(b + k) = h^2 + k^2 + 2gh + 2fy$

locus : $ax + by + ag + fb = x^2 + y^2 + gx + fy$

Q **Find the equation to the locus of the middle point of the chord of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which subtends right angle at a given point (a, b).**

Sol. for ΔAMN

P is circumcentre

$\Rightarrow AP = r$

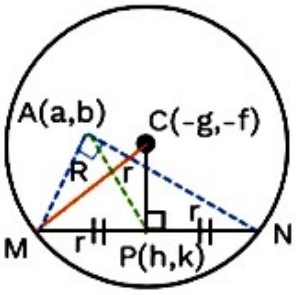
and $CM^2 = CP^2 + MP^2$

$R^2 = CP^2 + r^2$

$$g^2 + f^2 - c = (h + g)^2 + (k + f)^2 + (h - a)^2 + (k - b)^2$$

locus:

$$g^2 + f^2 - c = (x + g)^2 + (y + f)^2 + (x - a)^2 + (y - b)^2$$



Q **Tangents are drawn to a unit circle with centre at origin from every point on the line 2x + y = 4. Prove that**
(i) chord of contact passes through a fixed point

Sol. Circle S : $x^2 + y^2 = 1$
let point on given line P(h, k)
$\Rightarrow 2h + k = 4$ ....(1)
equation of chord of contact w.r.t. P
$xh + yk = 1$ .....(2)
by (1) and (2)

$$\frac{x}{2} = \frac{y}{1} = \frac{1}{4} \Rightarrow (x, y) \equiv \left(\frac{1}{2}, \frac{1}{4}\right)$$

$\therefore$ chord of contact passes through fixed point $\left(\frac{1}{2}, \frac{1}{4}\right)$

(ii) Equation to the locus of middle point of chord of contact.

Sol. Let mid point be P(h, k)
hence equation $T = S_1$
$xh + yk - 1 = h^2 + k^2 - 1$
$hx + ky = h^2 + k^2$ ... (1)
from above result if it is chord of contact
from a point on given line then it passes
through $\left(\frac{1}{2}, \frac{1}{4}\right)$ hence in (1)

$$\frac{h}{2} + \frac{k}{4} = h^2 + k^2$$

locus : $4x^2 + 4y^2 = 2x + y$

Q **Chord of contact of the tangent drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Prove that a, b, c are in G.P.**

Sol. $OA = a$

$OB = b$

$OM = c$

$AB = \sqrt{a^2 - b^2}$ ...(1)

in ΔOBM

$BM = \sqrt{b^2 - c^2}$

Now area of ΔOAB

$\frac{1}{2} OA \cdot BM = \frac{1}{2} OB \cdot AB$

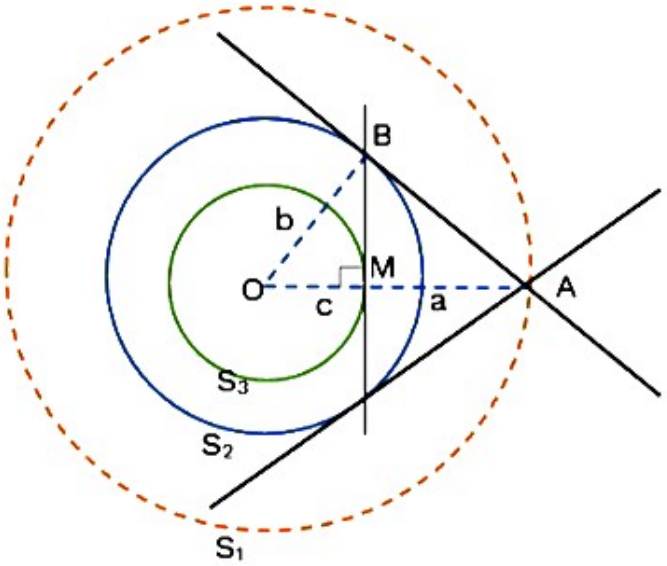
$a\sqrt{b^2 - c^2} = b\sqrt{a^2 - b^2}$

$a^2\left(b^2 - c^2\right) = b^2\left(a^2 - b^2\right)$

$\Rightarrow a^2c^2 = b^4$

$\Rightarrow b^2 = ac$

$\Rightarrow$ a, b, c in G.P.



Q **If the chord of contact of tangents drawn from P to the circle $x^2 + y^2 = a^2$ subtends a right angle at the centre, find the locus of P.**

Sol. Let P(h,k) and chord of contact is AB

given $\angle ACB = 90°$

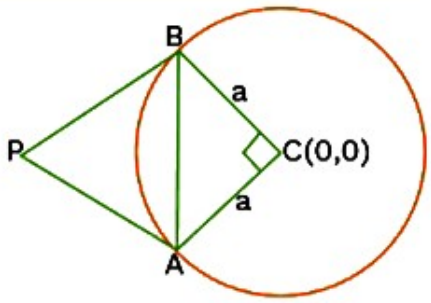
hence ACBP is square.

$PC = \sqrt{a^2 + a^2} = \sqrt{2}a$

$PC^2 = 2a^2$

$h^2 + k^2 = 2a^2$

locus : $x^2 + y^2 = 2a^2$

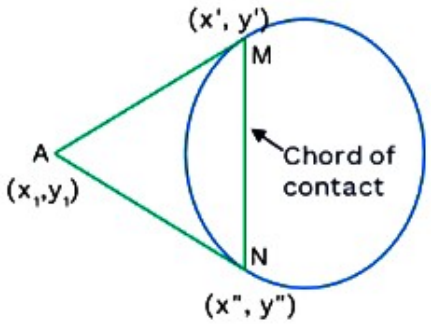


CHORD OF CONTACT

From any external point $A(x_1, y_1)$ draw pair of tangents AM and AN touching the circle at M(x', y') and N(x", y"), respectively. Then MN is the chord of contact with M, N as its points of contact.

Let the circle be $x^2 + y^2 = a^2$.

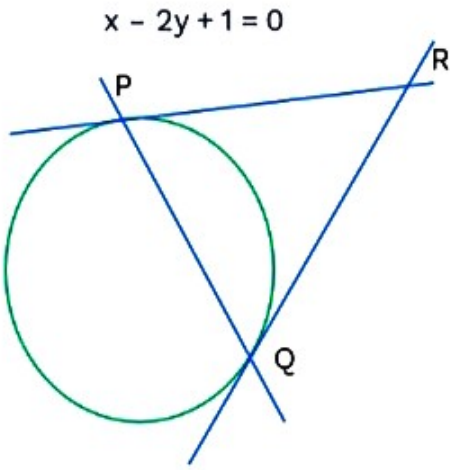
Then equation of tangents AM and AN are $xx' + yy' = a^2$ and $xx" + yy" = a^2$, respectively.



Since both tangents AM and AN pass through $A(x_1, y_1)$,
then $x_1x' + y_1y' = a^2$ and $x_1x'' + y_1y'' = a^2$.
Points $M(x', y')$ and $N(x'', y'')$ lie on $xx_1 + yy_1 = a^2$.
$\Rightarrow$ Equation of chord of contact MN is $xx_1 + yy_1 = a^2$.
or $T = 0$, where $T = xx_1 + yy_1 = a^2$
Equation of chord of contact at (x_1, y_1) with circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ or $T = 0$.

Q **If the straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ in points P and Q, then find the coordinates of the point of intersection of tangents drawn at P and Q to the circle $x^2 + y^2 = 25$.**

Sol. Let R(h, k) be the point of intersection of tangents drawn at P and Q to the given circle. Then PQ is the chord of contact of tangents drawn from R to $x^2 + y^2 = 25$.



So, its equation is

$$hx + ky - 25 = 0 \quad \ldots(i)$$

It is given that the equation of PQ is

$$x - 2y + 1 = 0 \quad \ldots(ii)$$

Since equation (i) and (ii) represent the same line, therefore

$$\frac{h}{1} = \frac{k}{-2} = \frac{-25}{1}$$

$\Rightarrow$ $h = -25$, $k = 50$

Hence, the required point is $(-25, 50)$.

Q **If the chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$, then prove that a, b, c are in G.P.**

Sol. Let (h, k) be a point on $x^2 + y^2 = a^2$. Then

$$h^2 + k^2 = a^2 \quad \ldots(i)$$

The equation of the chord of contact of tangents drawn from (h, k) to $x^2 + y^2 = b^2$ is

$$hx + ky = b^2 \quad \ldots(ii)$$

This touches the circle $x^2 + y^2 = c^2$. Therefore $\left|\frac{-b^2}{\sqrt{h^2+k^2}}\right| = c$

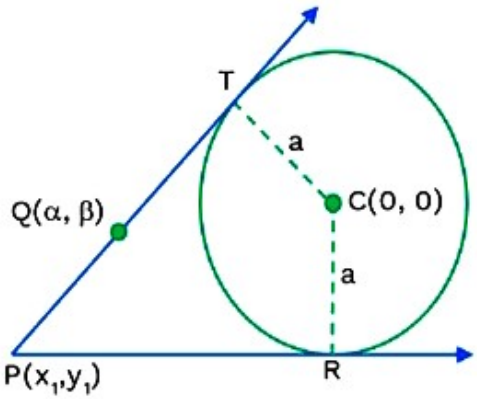
$\Rightarrow \left|\frac{-b^2}{\sqrt{a^2}}\right| = c$ [Using Eq. (i)]

$\Rightarrow b^2 = ac \Rightarrow$ a, b, c are in G.P.

PAIR OF TANGENTS

Let the circle be $x^2 + y^2 = a^2$

Its centre and radius are C(0, 0) and a, respectively. Let the given external point be $P(x_1, y_1)$.



From point $P(x_1, y_1)$, two tangents PT and PR can be drawn to the circle, touching circle at T and R, respectively.

Let $Q(\alpha, \beta)$ on PT, then equation of PQ is

$$y - y_1 = \frac{\beta - y_1}{\alpha - x_1}(x - x_1)$$

or $y(\alpha - x_1) - (\beta - y_1)x - \alpha y_1 + \beta x_1 = 0$

Length of perpendicular from C(0, 0) on Pt = a(radius)

$\Rightarrow \frac{|\beta x_1 - \alpha y_1|}{\sqrt{(\alpha - x_1)^2 + (\beta + y_1)^2}} = a$

or $(\beta x_1 - \alpha y_1)^2 = a^2\left\{(\alpha - x_1)^2 + (\beta - y_1)^2\right\}$

$\Rightarrow$ Locus of $Q(\alpha, \beta)$ is $(yx_1 - xy_1)^2 = a^2\left\{(x - x_1)^2 + (y - y_1)^2\right\}$

$\Rightarrow y^2x_1^2 - x^2y_1^2 - 2xyx_1y_1 = a^2\left\{x^2 + x_1^2 - 2xx_1 + y^2 + y_1^2 - 2yy_1\right\}$

$\Rightarrow y^2\left(x_1^2 + y_1^2 - a^2\right) + x^2\left(x_1^2 + y_1^2 - a^2\right) - a^2\left(x_1^2 + y_1^2 - a^2\right)$

$= (xx_1 + yy_1 - a^2)^2$

$\Rightarrow \left(x^2 + y^2 - a^2\right)\left(x_1^2 + y_1^2 - a^2\right) = \left(xx_1 + yy_1 - a^2\right)^2$

$\Rightarrow SS_1 = T^2$

Examples :

Q **Show that the equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(gx + fy)^2 = c(x^2 + y^2)$.**

Sol. $SS_1 = T^2$

$$(x^2 + y^2 + 2gx + 2fy + c)(c) = \{x(0) + y(0) + g(x + 0) + f(y + 0) + c)\}^2$$

$$(x^2 + y^2 + 2gx + 2fy + c)c = \{(gx + fy) + c\}^2$$

$$c(x^2 + y^2) + 2gcx + 2fcy + c^2 = (gx + fy)^2 + c^2 + 2c(gx + fy)$$

$$c(x^2 + y^2) = (gx + fy)^2$$

Hence proved.

Q **Tangents are drawn to the circle $x^2 + y^2 = a^2$ from two points on the axis of x equidistant from the point $(\alpha, 0)$. Show that the locus of their intersection is $\alpha y^2 = a^2(\alpha - x)$.**

Sol. Let point of intersection be P(h, k)

equation of pair of tangents from P

$SS_1 = T^2$

$$(x^2 + y^2 - a^2)(h^2 + k^2 - a^2) = (hx + ky - a^2)^2$$

for points A and B put y = 0

$$(x^2 - a^2)(h^2 + k^2 - a^2) = (hx - a^2)^2$$

$$= h^2x^2 + a^4 - 2a^2hx$$

$$(k^2 - a^2)x^2 + 2a^2hx - a^2(h^2 + k^2) = 0 \quad \rightarrow x_1, x_2$$

$$x_1 + x_2 = \frac{-2a^2h}{k^2 - a^2}$$

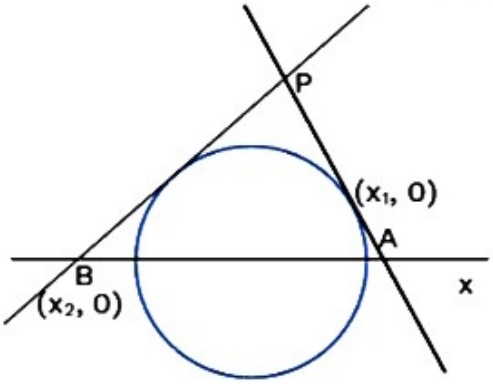
$\therefore$ mid point of AB is $(\alpha, 0)$

$$\Rightarrow \frac{x_1 + x_2}{2} = \alpha$$

$$\frac{-a^2h}{k^2 - a^2} = \alpha$$

$$\Rightarrow \alpha k^2 = a^2(\alpha - h)$$

Locus : $\alpha y^2 = a^2(\alpha - x)$



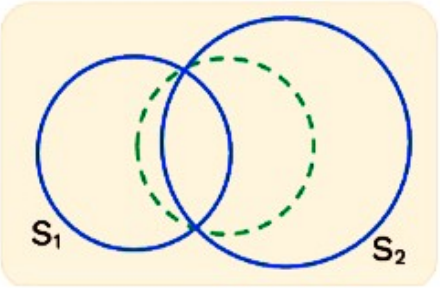
Hence proved.

Family of Circles :

Type–1 :

Equation of the family of circles which passes through the points of intersection of two circles.

$S_1 = 0$ and $S_2 = 0$ is $S_1 + \lambda S_2 = 0$; $\lambda \neq -1$



Example :

Q1 **Find the equation of a circle which passes through the point of intersection of $S_1 = 0$ and $S_2 = 0$**

$S_1 \equiv x^2 + y^2 - 4x + 6y - 3 = 0$

$S_2 \equiv x^2 + y^2 + 4x - 6y - 12 = 0$

(i) Which passes through (0,0)

Sol. Let circle be $S_1 + \lambda S_2 = 0$

$$\left(x^2 + y^2 - 4x + 6y - 3\right) + \lambda\left(x^2 + y^2 + 4x - 6y - 12\right) = 0 \quad \ldots(1)$$

on satisfying (0,0)

$$(-3) + \lambda(-12) = 0 \Rightarrow \frac{-1}{4} = \lambda$$

Put λ into (1)

$$\left(x^2 + y^2 - 4x + 6y - 3\right) - \frac{1}{4}\left(x^2 + y^2 + 4x - 6y - 12\right) = 0$$

$3x^2 + 3y^2 - 20x + 30y = 0$

(ii) Centre lies on x-axis or centre lies on y-axis.

Sol. From (1)

$$(1+\lambda)x^2 + (1+\lambda)y^2 + 4(\lambda - 1)x - 6(\lambda - 1)y - 3(1 + 4\lambda) = 0$$

If centre lies on x-axis then

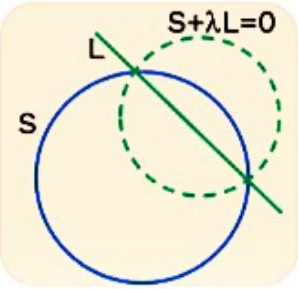
$$-6(\lambda - 1) = 0 \Rightarrow \lambda = 1$$

equation: $2x^2 + 2y^2 - 15 = 0$

$x^2 + y^2 = \frac{15}{2}$ (same for centre on y-axis)

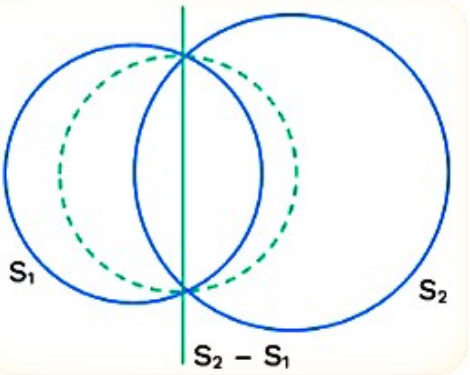
Type–2 :

Equation of the family of circles passes through the point of intersection of circles S=0 and a line L = 0 is given by $S + \lambda L = 0$



Modifying Type-1 using Type-2

$$S_1 + \lambda(S_2 - S_1) = 0$$



Examples :

Q **Find the equation of a circle drawn on the chord $x\cos\alpha + y\sin\alpha = p$ of the circle $x^2 + y^2 = a^2$ as its diameter.**

Sol. Equation of circles is

$(x^2 + y^2 - a^2) + \lambda(x\cos\alpha + y\sin\alpha - p) = 0$

$x^2 + y^2 + \lambda\cos\alpha \cdot x + \lambda\sin\alpha \cdot y - (\lambda p + a^2) = 0$

$\because$ line L is diameter of this circle hence centre lies on this line

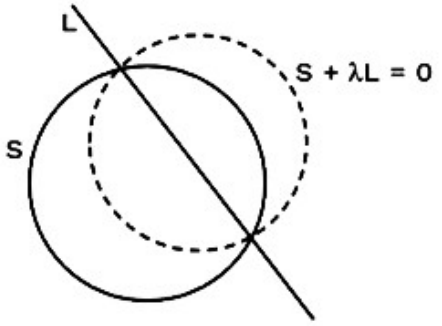
Centre : $\left(\frac{-\lambda}{2}\cos\alpha, \frac{-\lambda}{2}\sin\alpha\right)$

on satisfying the given line

$\left(\frac{-\lambda}{2}\cos\alpha\right)\cos\alpha + \left(\frac{-\lambda}{2}\sin\alpha\right)\sin\alpha = p$

$-\frac{\lambda}{2}(1) = p \Rightarrow \lambda = -2p$

Equation : $x^2 + y^2 - 2p\cos\alpha \cdot x - 2p\sin\alpha \cdot y + 2p^2 - a^2 = 0$



Q **Show that the equation $x^2 + y^2 - 2x + \lambda y - 8 = 0$ represents for different values of λ, a system of circles passing through two fixed points A and B on the x-axis, and also find the equation of that circle of the system the tangent to which at A and B meet on the line x + 2y + 5 = 0.**

Sol. Given equation of circle is $\left(x^2 + y^2 - 2x - 8\right) + \lambda\left(y\right) = 0$

which is of the form $S + \lambda L = 0$

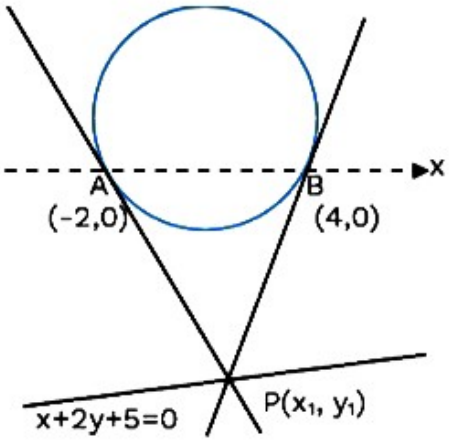
$S : x^2 + y^2 - 2x - 8 = 0$

$L : y = 0$

which always passes through point of intersection of S = 0 and L = 0

now intersection $x^2 + 0^2 - 2x - 8 = 0$

$(x - 4)(x + 2) = 0 \Rightarrow x = -2, 4$



points A(−2, 0) and B(4, 0)

Let $P(x_1, y_1)$

hence $x_1 + 2y_1 + 5 = 0$ .....(2)

now equation of chord of contact AB from P

$$xx_1 + yy_1 - \left(x + x_1\right) + \frac{\lambda}{2}\left(y + y_1\right) - 8 = 0$$

$$\left(x_1 - 1\right)x + \left(y_1 + \frac{\lambda}{2}\right)y - x_1 + \frac{\lambda}{2}y_1 - 8 = 0$$

on comparing it with y = 0

$$x_1 = 1, \quad x_1 - \frac{\lambda}{2}y_1 + 8 = 0$$

$$9 = \frac{\lambda}{2}y_1 \Rightarrow y_1 = \frac{18}{\lambda}$$

in (2),

$$1 + 2\left(\frac{18}{\lambda}\right) + 5 = 0 \Rightarrow 6 = \frac{-36}{\lambda} \Rightarrow \lambda = -6$$

Equation of circle : $x^2 + y^2 - 2x - 6y - 8 = 0$

Q **Find the equation of a circle which passes through the point of contact of the tangents drawn from the origin to the circle $x^2 + y^2 - 11x + 13y + 17 = 0$.**

Sol. Equation of chord of contact from origin

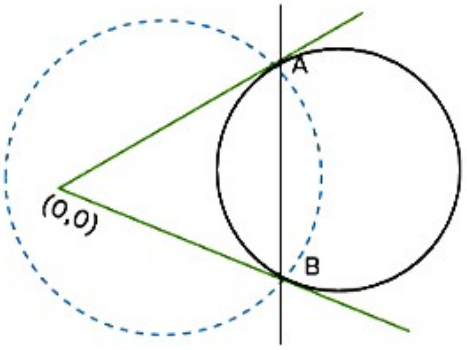
$$L : x(0) + y(0) - 11\left(\frac{x+0}{2}\right) + 13\left(\frac{y+0}{2}\right) + 17 = 0$$

$$L : \frac{-11x}{2} + \frac{13y}{2} + 17 = 0$$

$$L : 11x - 13y - 34 = 0$$

required family of circle is Õ ¾

$$\left(x^2 + y^2 - 11x + 13y + 17\right) + \lambda\left(11x - 13y - 34\right) = 0$$



Type-3 :

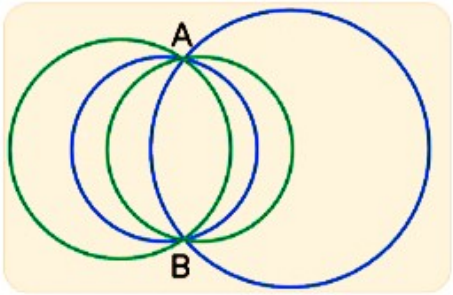
Equation of the family of circles passes through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$

Equation of circle passing through (x_1, y_1), (x_2, y_2) in diametrical form :

$S + \lambda L = 0$

$$S \equiv (x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

L ≡ line in two point form



Example :

Q **Find equation of circumcircle of Δ whose vertices are (1, 0), (2, 0), (3, 1)**

Sol. Let A(1, 0), B(2, 0), C(3,1)

Equation of circle passing through A and B

$(x-1)(x-2) + y^2 + \lambda(y) = 0$

which also passes through C hence.

$(3-1)(3-2) + 1^2 + \lambda(1) = 0 \Rightarrow \lambda = -3$

Equation: $(x-1)(x-2) + y^2 - 3y = 0$

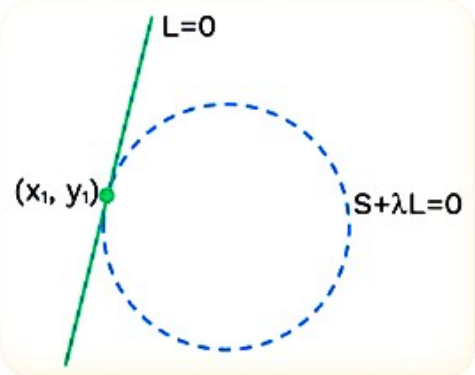
$x^2 + y^2 - 3x - 3y + 2 = 0$

Type–4 (Point circle)

Equation of family of circles touching a line

L = 0 at its fixed point (x_1, y_1) is

$$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$$



Examples :

Q **Find the equation of a circle which touches the line 2x–y=4 at the point (1, –2) and passes through (3, 4).**

Sol. Equation: $(x-1)^2+(y+2)^2+\lambda(2x-y-4)=0$

also it passes through (3, 4) hence

$2^2+6^2+\lambda(6-4-4)=0$

$2\lambda=40 \Rightarrow \lambda=20$

equation : $(x-1)^2+(y+2)^2+20(2x-y-4)=0$

$x^2+y^2+38x-16y-75=0$

Q **Find the equation of the circle which passes through the point (-1, 2) and touches the circle $x^2+y^2-8x+6y=0$ at origin.**

Sol. Equation of tangent at (0,0)

$x(0)+y(0)-4(x+0)+3(y+0)=0$

$L: 4x-3y=0$

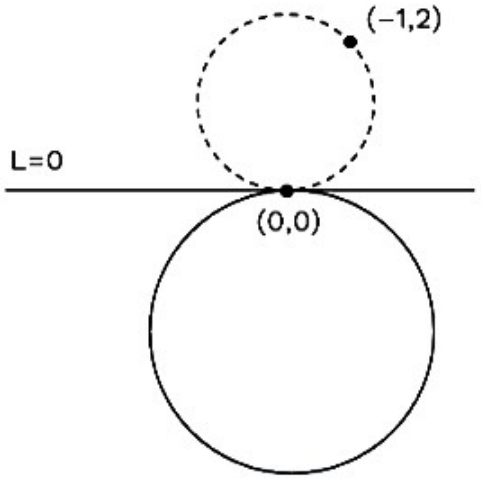
Equation of required circle:

$x^2+y^2+\lambda(4x-3y)=0$ ...(1)

which passes through (–1, 2)

$1+4+\lambda(-4-6)=0 \Rightarrow \lambda=\frac{1}{2}$

Equation : $x^2+y^2+2x-\frac{3}{2}y=0$



Q **Find equation of circles to which line 4x + 3y = 10 is a common tangent at (1, 2) and radius of each circle is 5.**

Sol. Equation is $(x-1)^2+(y-2)^2+\lambda(4x+3y-10)=0$

$\Rightarrow x^2+y^2+(4\lambda-2)x+(3\lambda-4)y+(5-10\lambda)=0$

radius r = 5

$r^2 = 25$

$$\left(\frac{4\lambda - 2}{2}\right)^2 + \left(\frac{3\lambda - 4}{2}\right)^2 - (5 - 10\lambda) = 25$$

$$\frac{25\lambda^2}{4} = 25 \Rightarrow \lambda = \pm 2$$

$\lambda = 2 \Rightarrow x^2 + y^2 + 6x + 2y - 15 = 0$
$\lambda = -2 \Rightarrow x^2 + y^2 - 10x - 10y + 25 = 0$

Q **The line $2x - 3y + 1 = 0$ is tangent to a circle $S = 0$ at $(1, 1)$. If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S.**

Sol. Equation is

$$(x-1)^2 + (y-1)^2 + \lambda(2x - 3y + 1) = 0$$

$$\Rightarrow x^2 + y^2 + (2\lambda - 2)x - (2 + 3\lambda)y + (\lambda + 2) = 0$$

$r = \sqrt{13} \Rightarrow r^2 = 13$

$$\left(\frac{2\lambda - 2}{2}\right)^2 + \left(\frac{2 + 3\lambda}{2}\right)^2 - (\lambda + 2) = 13$$

$13\lambda^2 = 13 \times 4 \Rightarrow \lambda = \pm 2$

$\lambda = 2 \Rightarrow x^2 + y^2 + 2x - 8y + 4 = 0$

$\lambda = -2 \Rightarrow x^2 + y^2 - 6x + 4y = 0$

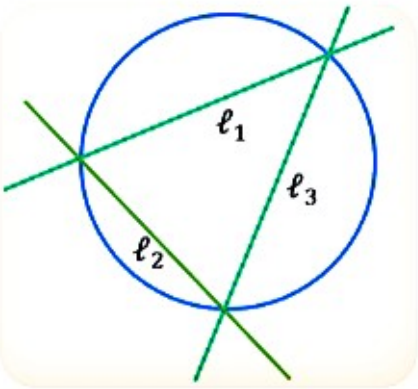
Type–5 :

Equation of circle passing through points of intersection of lines

$$\ell_1 = 0, \ell_2 = 0, \ell_3 = 0$$

(circumcircle of the triangle thus formed) is given by

$$\ell_1\ell_2 + \lambda\ell_2\ell_3 + \mu\ell_3\ell_1 = 0$$



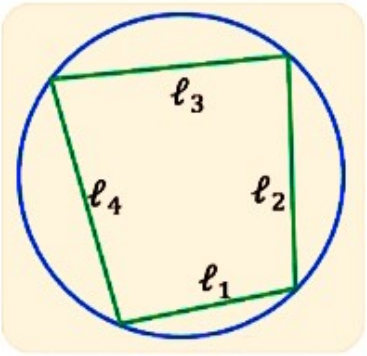
Note :

To find λ and μ
coefficient of x^2 = coefficient of y^2 and
coefficient of $xy = 0$

Type–6 :

Equation of a circle circumscribing a quadrilateral whose sides in order are represented the line $\ell_1 = 0; \ell_2 = 0; \ell_3 = 0;\ \ell_4 = 0$ is given by

$$\ell_1\ell_3 + \lambda\ell_2\ell_4 = 0$$



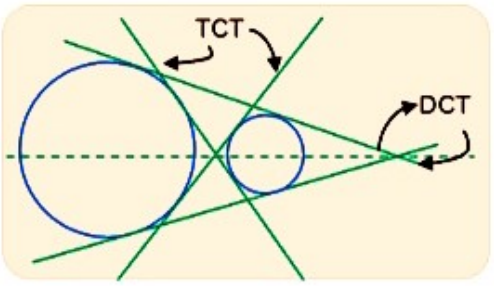
Note :

To find λ

coefficient of x^2 = coefficient of y^2 and coefficient of $xy = 0$

Common tangents

1. Direct common Tangent (DCT) (External Common Tangent)
2. Transverse Common Tangent (TCT) (Internal Common Tangent)

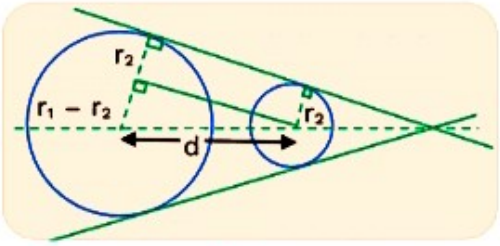


Direct Common Tangent

The centre of both the circles lie on the same side of the tangent line.

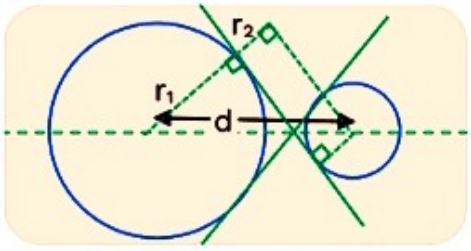
Transverse Common Tangent

The centre of both the circles lie on the opposite side of the tangent line.

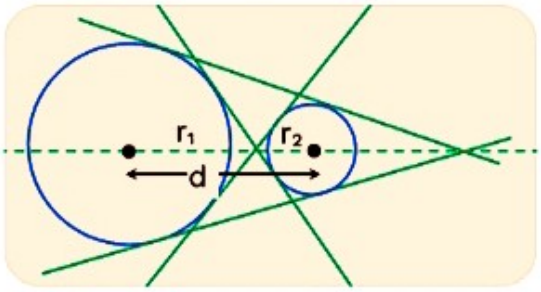


Length of DCT/TCT

$$L_{ext} = \sqrt{d^2 - (r_1 - r_2)^2}$$



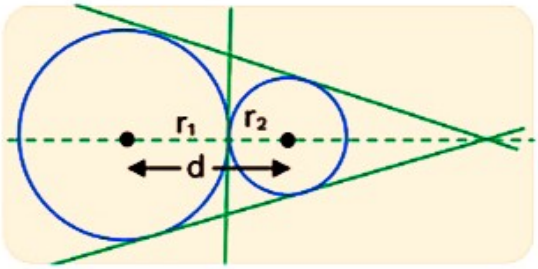
$$L_{int} = \sqrt{d^2 - (r_1 + r_2)^2}$$



Position of circles
(And Number of Common Tangents)
Equation of DCT/TCT

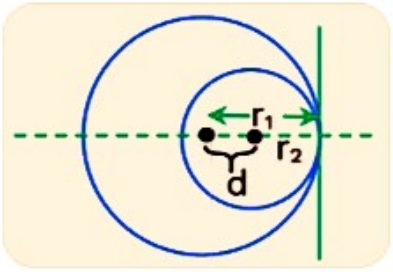
(1) If two circles are separated then $d > r_1 + r_2$

4 common tangents = $\begin{cases} 2DCT \\ 2TCT \end{cases}$



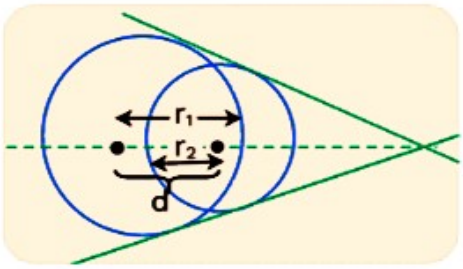
(2) If 2 circles touch externally then $d = r_1 + r_2$

3 common tangents = $\begin{cases} 2DCT \\ 1TCT \end{cases}$



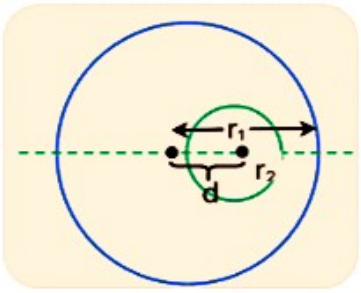
(3) If 2 circles touches internally then $d = |r_1 - r_2|$

One common tangent (DCT)



(4) If 2 circles intersect each other then $|r_1 - r_2| < d < r_1 + r_2$

Two common tangents (2 DCT)



(5) If one circle lies completely inside of another circle then $d < |r_1 - r_2|$

No common tangent.

Examples :

Q **Find the range of 'r' so that the circles $(x-1)^2 + (y-3)^2 = r^2$ and $(x-4)^2 + (y-1)^2 = 9$ intersects at 2 distinct points.**

Sol. $C_1(1,3)$, $r_1 = |r|$

$C_2(4,1)$, $r_2 = 3$

$d = C_1C_2 = \sqrt{9+4} = \sqrt{13}$

$|r_1 - r_2| < d < r_1 + r_2$

$||r| - 3| < \sqrt{13} < |r| + 3$

$\Rightarrow -\sqrt{13} < |r| - 3 < \sqrt{13} \Rightarrow 3 - \sqrt{13} < |r| < 3 + \sqrt{13}$

$-\left(3+\sqrt{13}\right) < r < \left(3+\sqrt{13}\right)$ ...(1)

$\Rightarrow \sqrt{13} < |r| + 3 \Rightarrow |r| > \sqrt{13} - 3$

$r \in \left(-\infty, 3-\sqrt{13}\right) \cup \left(\sqrt{13}-3, \infty\right)$ ...(2)

(1) $\cap$ (2)

$r \in \left(-\left(3+\sqrt{13}\right), 3-\sqrt{13}\right) \cup \left(\sqrt{13}-3, 3+\sqrt{13}\right)$

Q. **A circle is given by $x^2 + (y-1)^2 = 1$, another circle C touches it externally and also the x-axis, then the locus of its centre is**

(A) $\{(x,y) : x^2 = 4y\} \cup \{(x,y) : y \le 0\}$ **(B)** $\{(x,y) : x^2 + (y-1)^2 = 4\} \cup \{(x,y) : y \le 0\}$

(C) $\{(x,y) : x^2 = y\} \cup \{(0,y) : y \le 0\}$ **(D)** $\{(x,y) : x^2 = 4y\} \cup \{(0,y) : y \le 0\}$

Sol. **(D)**

$C_1(0, 1)$, $r_1 = 1$

Let $C_2(x, y)$, $r_2 = |y|$

Now circles touch externally then

$C_1C_2 = r_1 + r_2$

$\sqrt{x^2 + (y-1)^2} = 1 + |y|$

on squaring

$x^2 + y^2 - 2y + 1 = 1 + y^2 + 2|y|$

$x^2 - 2y = 2|y|$

for $y > 0 \Rightarrow x^2 = 4y$

for $y \le 0 \Rightarrow x^2 - 2y = -2y \Rightarrow x = 0$

Q **Find common tangent to the circles $x^2 + y^2 = 1$ and $(x-1)^2 + (y-3)^2 = 4$.**

Sol. $C_1(0, 0)$ $r_1 = 1$

$C_2(1, 3)$ $r_2 = 2$

$C_1C_2 = \sqrt{1+9} = \sqrt{10} > r_1 + r_2$

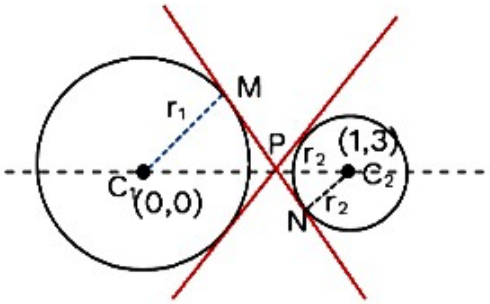
circles are separated $\Rightarrow$ 4 common tangents

TCT :

$\Delta C_1MP \sim \Delta C_2NP$

$\therefore C_1P : C_2P = r_1 : r_2 = 1 : 2$

$P\left(\frac{1}{3}, \frac{3}{3}\right) \equiv P\left(\frac{1}{3}, 1\right)$



Let tangent

$(y-1) = m\left(x - \frac{1}{3}\right)$

$\Rightarrow mx - y + 1 - \frac{m}{3} = 0$ ...(1)

now $p = r_1$

$$\left|\frac{1-\frac{m}{3}}{\sqrt{m^2+1}}\right| = 1 \Rightarrow \left(1-\frac{m}{3}\right)^2 = m^2 + 1$$

$$1 + \frac{m^2}{9} - \frac{2m}{3} = m^2 + 1 \Rightarrow \frac{8}{9}m^2 = \frac{-2m}{3}$$

$$m = 0, m = \frac{-3}{4}$$

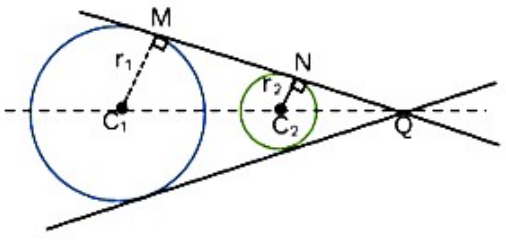
in (1), $m = 0 \Rightarrow y = 1$

$m = \frac{-3}{4} \Rightarrow \frac{-3}{4}x - y + 1 + \frac{1}{4} = 0$

$3x + 4y = 5$

DCT:

$\Delta C_1MQ \sim \Delta C_1NQ$



Q divides C_1C_2 in $r_1 : r_2$ externally

Q(−1, −3)

Let tangent be

$(y+3) = m(x+1) \Rightarrow mx - y + m - 3 = 0$ ...(2)

now $p = r_1$

$$\left|\frac{m-3}{\sqrt{m^2+1}}\right| = 1 \Rightarrow (m-3)^2 = m^2 + 1$$

$m^2 - 6m + 9 = m^2 + 1$

$\Rightarrow m = \frac{4}{3}, \quad m = \text{N.D.}$

equation (from 2)

for $m = \frac{4}{3} \Rightarrow \frac{4}{3}x - y + \frac{4}{3} - 3 = 0$

$4x - 3y - 5 = 0$

for $m = \text{N.D.} \quad \Rightarrow x = -1$

Radical Axis

Radical axis of 2 circles is the locus of a point whose powers w.r.t. the two circles are equal. The equation of radical axis of two circles $S_1 = 0$ and $S_2 = 0$ is given by $S_1 - S_2 = 0$

Note :

(a) If two circles intersect, then the radical axis is the common chord of the two circles.

(b) If two circles touch each other then the radical axis is the common tangent of the two circles at the common point of contact.

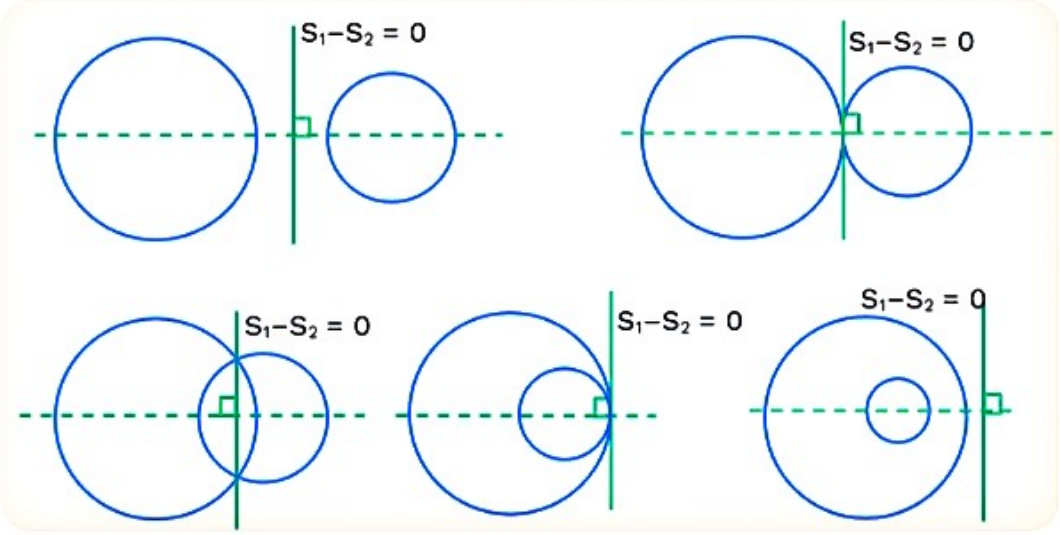
(c) Radical axis is always perpendicular to the line joining the centres of the two circles.

(d) Radical axis need not always pass through the mid point of the line joining the centres of the two circles.

(e) Radical axis bisects a common tangent between the two circles.

(f) If one circle is contained in another circle when radical axis passes outside to both the circles.

(g) Concentric circles do not have radical axis.



Examples :

Q **Show that the equation of a straight line meeting the circle $x^2 + y^2 = a^2$ in 2 points at equal distance 'd' from the point (x_1, y_1) on its circumference is**

$$xx_1 + yy_1 - a^2 + \frac{d^2}{2} = 0$$

Sol.

$S_1 : x^{\delta} + y = a$

$S_2 : (x - x_1)^2 + (y - y_1)^2 = d^2$

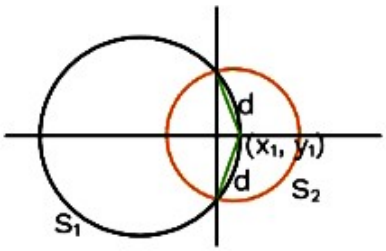
Equation of line: $S_1 - S_2 = 0$

$2xx_1 - x_1^2 + 2yy_1 - y_1^2 = a^2 - d^2$

$2xx_1 + 2yy_1 = (x_1^2 + y_1^2 + a^2) - d^2$

$2xx_1 + 2yy_1 = 2a^2 - d^2 \qquad (\because x_1^2 + y_1^2 = a^2)$

$\Rightarrow xx_1 + yy_1 - a^2 + \frac{d^2}{2} = 0$



Q **Prove that the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ will bisect circumference of the circle $x^2 + y^2 + 2g'x + 2f'y + c' = 0$ if $2g'(g - g') + 2f'(f - f') = c - c'$**

Sol.

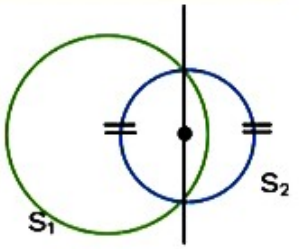
Common chord of $S_1 = 0$ and $S_2 = 0$ will be diameter of $S_2 = 0$

Common chord, $S_1 - S_2 = 0$

$2(g - g')x + 2(f - f')y + c - c' = 0$

Now put centre $C_2(-g', -f')$ in it

$2g'(g - g') + 2f'(f - f') = c - c'$



Q **Tangents are drawn to the circle $x^2 + y^2 = 12$ at the points where it is met by the circle $x^2 + y^2 - 5x + 3y - 2 = 0$. Find the point of intersection of tangents.**

Sol.

Let $P(x_1, y_1)$

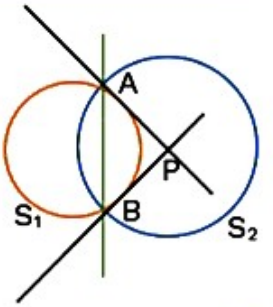
AB is chord of contact from P on S_1

Hence equation

$xx_1 + yy_1 = 12$ ...(1)

Also AB is common chord

hence equation $S_1 - S_2 = 0$



$5x - 3y = 10$ ...(2)

by (1) and (2)

$$\frac{x_1}{5} = \frac{y_1}{-3} = \frac{12}{10} \Rightarrow P\left(6, -\frac{18}{5}\right)$$

Q **Find the equation of a circle which bisects the circumference of the circles $x^2 + y^2 = 1$, $x^2 + y^2 + 2x = 3$ and $x^2 + y^2 + 2y = 3$.**

Sol. Let the circle is

$S : x^2 + y^2 + 2gx + 2fy + c = 0$

S and S_1 have common chord as diameter of S_1

$\Rightarrow S - S_1 = 0$

$2gx + 2fy + c + 1 = 0$

on satisfying, $C_1(0, 0) \Rightarrow c = -1$ ...(1)

S and S_2 have common chord as diameter of S_2

$\Rightarrow S - S_2 = 0$

$2gx - 2x + 2fy + c + 3 = 0$

on satisfying $C_2(-1, 0) \Rightarrow -2g + 2 + 2 = 0$

$g = 2$ ...(2)

S and S_3 have common chord as diameter of S_3

$\Rightarrow S - S_3 = 0$

$2gx + 2fy - 2y + c + 3 = 0$

on satisfying, $C_1(0, -1) \Rightarrow -2f + 2 + 2 = 0$

$f = 2$ ...(3)

equation of circle

$S : x^2 + y^2 + 4x + 4y - 1 = 0$

Q **Find the locus of the centre of circles which bisect the circumference of the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 2x + 6y + 1 = 0$**

Sol. Let centre be C(h, k)

equation $S \equiv x^2 + y^2 - 2hx - 2ky + c = 0$

$S - S_1 = 0 \Rightarrow -2hx - 2ky + c + 4 = 0$

it is diameter of $S_1 = 0$ hence

$c + 4 = 0 \Rightarrow c = -4$

$S - S_2 = 0 \Rightarrow 2x - 2hx - 6y - 2ky + c - 1 = 0$

it is diameter of $S_2 = 0$ hence

$2 - 2h + 18 + 6k - 5 = 0$

$2h - 6k = 15$

locus : $2x - 6y = 15$

Q **Find the equation of the circle which bisects the circumference of the circle $x^2 + y^2 + 2y - 3 = 0$ and touches the line $x - y = 0$ at origin.**

Sol. Equation of circle is Õ Ñ

$S_1 : x^2 + y^2 + \lambda(x - y) = 0$

Also given circle $S_2 : x^2 + y^2 + 2y - 3 = 0$

$S_1 - S_2 = 0 \Rightarrow \lambda x - \lambda y - 2y + 3 = 0$

which is diameter of $S_2 = 0$ hence

$\lambda(0) - \lambda(-1) - 2(-1) + 3 = 0$

$\lambda = -5$

Equation : $x^2 + y^2 - 5x + 5y = 0$

Radical centre :

The radical axis of three circles, taken in pairs, meet at a point, which is called their radical centre.

Radical axis of S_1 and S_2 : $S_1 - S_2 = 0$ ...(i)

S_2 and S_3 : $S_2 - S_3 = 0$ ...(ii)

S_3 and S_1 : $S_3 - S_1 = 0$ ...(iii)

Formally of lines passing through (i) and (ii)

$(S_1 - S_2) + \lambda (S_2 - S_3) = 0$

when $\lambda = 1$ we get $S_1 - S_3 = 0$ which is 3rd radical axis. Hence 3rd R.A. also passes through O. O is radical centre.

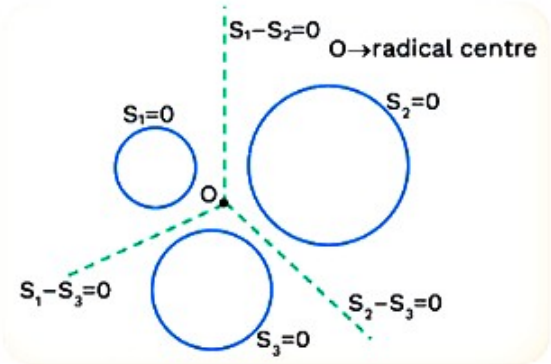
Discriminant of (i), (ii), (iii)

$$D = \begin{vmatrix} g_1 - g_2 & f_1 - f_2 & c_1 - c_2 \\ g_2 - g_3 & f_2 - f_3 & c_2 - c_3 \\ g_3 - g_1 & f_3 - f_1 & c_3 - c_1 \end{vmatrix} = 0 \ ;$$

use $R \to R_1 + R_2 + R_3$

$\Rightarrow D = 0$

Note: Solve any 2 radical axis to get radical centre from which tangent to three circle are equal.



Note :

(i) Radical axis taken 2 at a time will be concurrent at a point.

(ii) Radical centre of three circles described on sides of a Δ as diameter is orthocentre of the Δ.

(iii) Radical centre exists only for three circles.

(iv) The radical centre can be found by solving the equations $S_1 = S_2 = S_3 = 0$

(v) If two circles cut a third circle orthogonally, then the radical axis of the two circles will pass through the center of the third circle.

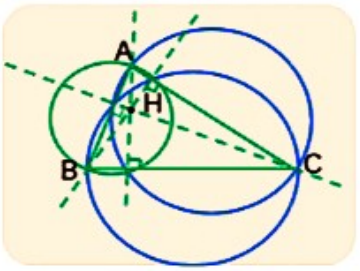
OR

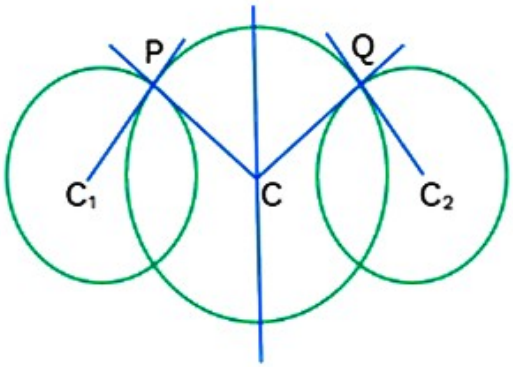
The locus of the center of the circle cutting two given circles orthogonally is the radical axis of the two given circles.

$C_1P \perp CP$ and $C_2Q \perp CQ$

But CP = CQ = Radius

Hence C lies on the radical axis of S_1 and S_2.





Q **Lines $lx + my + n = 0$; $px + qy + r = 0$ intersect the circle $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ at 4 points A, B, C and D respectively. If these points are concyclic then prove that** $\begin{vmatrix} 2(g_1 - g_2) & 2(f_1 - f_2) & c_1 - c_2 \\ l & m & n \\ p & q & r \end{vmatrix} = 0$.

Sol. $PT^2 = PA \cdot PB$

$PQ2 = PC \cdot PD$

$\because$ ABDC concyclic

So, $PA \cdot PB = PC \cdot PD$

Hence, $PT^2 = PQ^2$

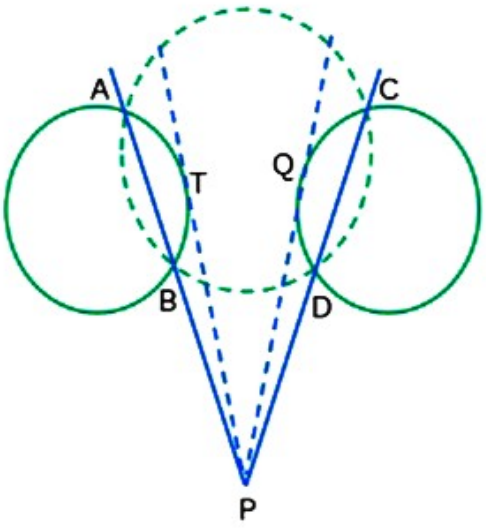
$PT = PQ$

Hence P lies on the radical axis of these two circles R.A. :

$2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0$

R.A. and the two given lines are concurrent at the point P.

So $\begin{vmatrix} 2(g_1 - g_2) & 2(f_1 - f_2) & c_1 - c_2 \\ l & m & n \\ p & q & r \end{vmatrix} = 0$



Coaxial system of circles :

Definition

A system of circles, every 2 of which have the same radical axis, is called coaxial system of circles.

Examples :

Q **Find the equation of the circle passes through (1, 1) belonging to the system of coaxial circles which touches $x^2 + y^2 = 8$ at (2, 2).**

Sol. Equation of tangent of given circle at (2, 2) is

$2x + 2y = 8 \Rightarrow x + y = 4$

Now circle touching this line at (2, 2) is

$$(x-2)^2 + (y-2)^2 + \lambda(x + y - 4) = 0$$

on satisfying this by point (1, 1)

$1 + 1 + \lambda(-2) = 0 \Rightarrow \lambda = 1$

circle : $(x-2)^2 + (y-2)^2 + (x + y - 4) = 0$

$$x^2 + y^2 - 3x - 3y + 4 = 0$$

Q **From a point P tangents drawn to the circles**

$$x^2 + y^2 + x - 3 = 0$$

$$3x^2 + 3y^2 - 5x + 3y = 0$$

and $4x^2 + 4y^2 + 8x + 7y + 9 = 0$

are of equal length. Find the equation of the circle which passes through P and which touches the line x + y = 5 at (6, –1).

Sol. P will be radical centre hence

$S_1 : x^2 + y^2 + x - 3 = 0$

$$S_2 : x^2 + y^2 - \frac{5}{3}x + y = 0$$

$$S_3 : x^2 + y^2 + 2x + \frac{7}{4}y + \frac{9}{4} = 0$$

Now radical axis

$$S_1 - S_2 = 0 \Rightarrow \frac{8}{3}x - y - 3 = 0$$

$8x - 3y = 9$ ...(1)

$S_1 - S_3 = 0 \Rightarrow -x - \frac{7}{4}y - 3 - \frac{9}{4} = 0$

$4x + 7y + 21 = 0$ ...(2)

(2) × 2 – (1)

$\Rightarrow 17y + 51$

$\Rightarrow y = -3, x = 0$

$\Rightarrow P(0, -3)$

equation of required circle:

$(x-6)^2 + (y+1)^2 + \lambda(x + y - 5) = 0$

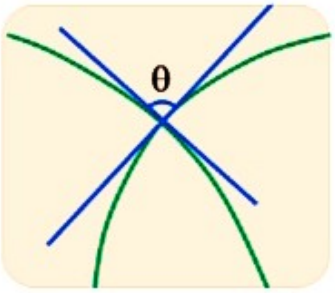
which passes through P(0, –3)

$\Rightarrow 36 + 4 + \lambda(-8) = 0 \Rightarrow \lambda = 5$

equation : $x^2 + y^2 - 7x + 7y + 12 = 0$

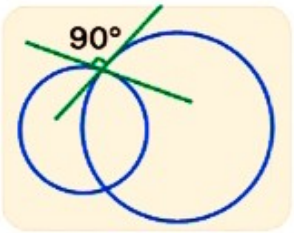
Angle between two curves :

Angle between two curves is the angle between the tangents at the point of intersection.



Orthogonality of two circles :

Two curves are said to be orthogonal if angle between them is 90° at point of intersection.



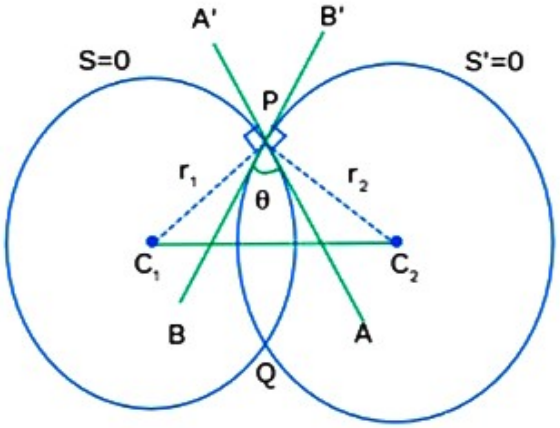
Note :

Line is tangent to itself.

ANGLE OF INTERSECTION OF TWO CIRCLES

Let the two circles $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and $S' \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ intersect each other at the point P and Q. The angle θ between two circles S = 0 and S' = 0 is defined as the angle between the tangents to the two circles at the point of intersection. θ must be taken acute angle.

C_1 and C_2 are the centres of circles S = 0 and S' = 0, then $C_1 \equiv (-g, -f)$ and $C_2 \equiv (-g_1, -f_1)$ and radii of circles S = 0 and S' = 0 are $r_1 = \sqrt{g^2 + f^2 - c}$ and $r_2 = \sqrt{g_1^2 + f_1^2 - c_1}$



Let $d = |C_1C_2|$ = Distance between their centres

$= \sqrt{(-g + g_1)^2 + (-f + f_1)^2}$

$$= \sqrt{\left(g^2 + f^2 + g_1^2 + f_1^2 - 2gg_1 - 2ff_1\right)}$$

$C_1P \perp AA' \Rightarrow \angle C_1PB = \frac{\pi}{2} - \theta$

$C_2P \perp BB' \Rightarrow \angle C_2PA = \frac{\pi}{2} - \theta$

$\Rightarrow \angle C_1PC_2 = \frac{\pi}{2} - \theta + \frac{\pi}{2} - \theta + \theta = \pi - \theta$

Now, in $\Delta PC_1C_2, \cos\theta = \left|\frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right|$ ...(i)

Q **Find the angle at which the circles $x^2 + y^2 + x + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect.**

Sol. The angle of intersection of two circles is given by

$$\cos\theta = \frac{r_1^2 + r_2^2 - C_1C_2}{2r_1r_2},$$

where r_1, r_2 are radii of two circles and C_1C_2 is the distance between their centres.

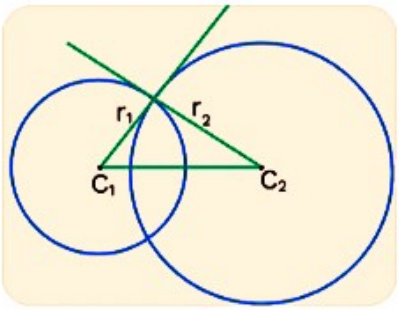
Here, $r_1 = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}} = r_1$ and $C_1C_2 = 1$

$\Rightarrow \cos\theta = 0$

$\Rightarrow \theta = \frac{\pi}{2}$

Condition of orthogonality of two circles :

$2g_1g_2 + 2f_1f_2 = c_1 + c_2$



Proof:

Let circles

$S_1 : x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$S_2 : x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$

Now $r_1^2 + r_2^2 = (C_1C_2)^2$

$(g_1^2 + f_1^2 - c_1) + (g_2^2 + f_2^2 - c_2) = (g_1 - g_2)^2 + (f_1 - f_2)^2$

$\Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

How to find circle orthogonal to 3 given circles:

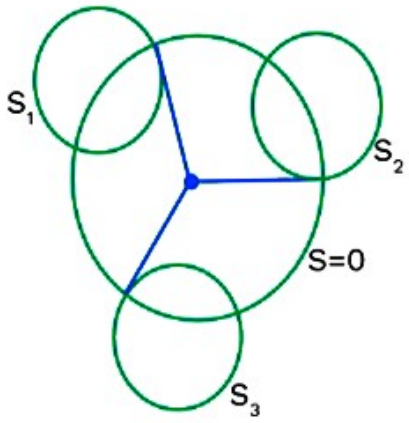
Method-I

Let the equation of circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

orthogonal to $S_1 = 0$, $S_2 = 0$, $S_3 = 0$

Get three relations in g, f, c and solve.



Method-II

Circle orthogonal to three given circles is one such circle whose center is the radical center of 3 given circles and radius equal to the length of tangent drawn from radical center to any of the three given circles.

Explanation:

Solve any two radical axis to get radical center and then compute $r = L_T$

Examples :

Q **If a circle intersects the circle $x^2 + y^2 + 2x + 4y + 1 = 0$ and lines $x = 0$, $y = 0$ orthogonally. Find the equation of circle.**

Sol. Let circle be

$S_1 : x^2 + y^2 + 2gx + 2fy + c = 0$

Now S_1 and given circle S_2 are orthogonal hence

$2g(1) + 2f(2) = c + 1$

$2g + 4f = c + 1$ ....(1)

Now $S_1 = 0$ is orthogonal to $x = 0 \Rightarrow x = 0$ is diameter

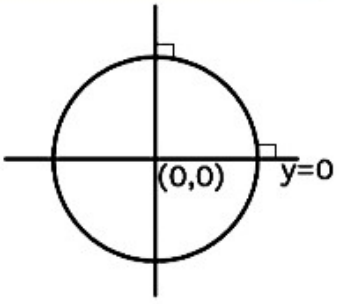
$S_1 = 0$ is orthogonal to $y = 0 \Rightarrow y = 0$ is diameter

$\Rightarrow$ centre: (0, 0)

hence $g = f = 0$

then $c = -1$

equation : $x^2 + y^2 = 1$



Q **Find the equation of the circle which passes through the origin, meets the x-axis orthogonally and cuts the circle $x^2 + y^2 = a^2$ at an angle of 45°.**

Sol. Clearly x-axis is a diameter

Let circle

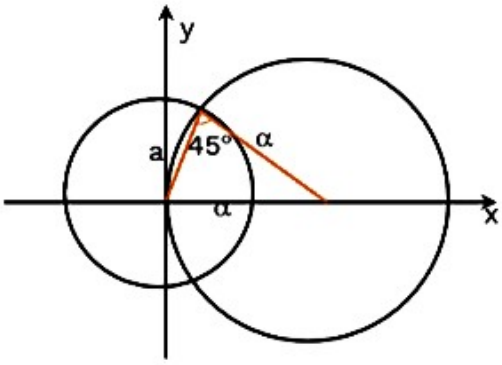
$S : (x-\alpha)^2 + y^2 = \alpha^2$

$\pm\cos 45^\circ = \dfrac{a^2+\alpha^2-\alpha^2}{2a\alpha}$

$\pm\dfrac{1}{\sqrt{2}} = \dfrac{a^2}{2a\alpha} \Rightarrow \alpha = \pm\dfrac{a}{\sqrt{2}}$

equation : $x^2 + y^2 - 2\alpha x = 0$

$x^2 + y^2 \pm \sqrt{2}ax = 0$



Q **Prove that locus of the centre of a variable circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which cuts the 2 given circles $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ orthogonally is the radical axis of 2 given circles.**

Sol. First and second circle are orthogonal, hence

$2gg_1 + 2ff_1 = c + c_1$ ...(1)

Also first and third circle are orthogonal hence

$2gg_2 + 2ff_2 = c + c_2$ ...(2)

Subtracting (2)from (1)

$2g(g_1 - g_2) + 2f(f_1 - f_2) = c_1 - c_2$

for locus put $g = -x$, $f = -y$

$\Rightarrow -2x(g_1 - g_2) - 2y(f_1 - f_2) = c_1 - c_2$

$\Rightarrow 2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0$

which is radical axis of circles.

Q **If the circles** $S_1 : x^2 + y^2 + 2x + 2ky + 6 = 0$

and $S_2 : x^2 + y^2 + 2kx + k = 0$

intersects orthogonally then find k

Sol. $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

$\Rightarrow 2(k) + 2k(0) = 6 + k$

$\Rightarrow k = 6$

Q **Show that the locus of the centres of a circle which cuts two given circles orthogonally is a straight line and hence deduce the locus of the centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 5x + 4y + 2=0$ orthogonally. Interpret the locus.**

Sol. Clearly locus of centre will be radical axis of the two circles which we have proved earlier.
For two given circles, locus of the centre of orthogonal circle is
$S_1 - S_2 = 0$
$\Rightarrow 9x - 10y + 7 = 0$

Q **Let $S_1 = x^2 - 10x + y^2 + 16 = 0$, and S_2 be the image of S_1 w.r.t the line $x - y = 0$ and S_3 be the image of S_2 w.r.t. $y = 0$. Let S' is the circle which cut all three circles orthogonally and S" is the circle of minimum radius which contain all three circles then which of the following is false.**
(A) equation of circle S' is $x^2 + y^2 = 16$
(B) equation of circle S' is $x^2 + y^2 = 64$
(C) area of circle S' is 16π
(D) area of circle S' is 36π

Sol. **(D)**

S' is the radical circle of S_1, S_2 and S_3

$\therefore$ It's center is the point of intersection of lines $x - y = 0$ and $y = 0$ i.e., (0, 0) and it's radius is length of tangent from (0, 0) to $S_1 \Rightarrow$ Radius of $S_1 = 4$.

$\therefore$ Equation of S_1 is $x^2 + y^2 = 16$

$\therefore$ Now, S" is the circle which contains all the circle S_1, S_2 and S_3.

$\therefore$ It's center is their radical center i.e., (0, 0) and R = maximum distance of origin from S_1

$= PC + R = 5 + 3 = 8$

$\Rightarrow$ Equation of S" is $x^2 + y^2 = 64$.